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Macquarie Coal Complex Transformation Precinct Master Plan

July 2026



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Acknowledgement of Country

The Department of Planning, Housing and Infrastructure acknowledges that it stands on Aboriginal land. We acknowledge the Traditional Custodians of the land, and we show our respect for Elders past, present and emerging through thoughtful and collaborative approaches to our work, seeking to demonstrate our ongoing commitment to providing places in which Aboriginal people are included socially, culturally and economically.

Artwork by Kyara Fernando Dunghutti (Dhanggati)

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**LAKE
MACQUARIE
CITY**

GLENCORE



Draft Macquarie Coal Complex Transformation Precinct Master Plan

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Image on front cover: Indicative visualisation of light industrial development within the Precinct.

Credit: GHD, 2026

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1 Introduction

Citation and commencement

This Document is the *draft Macquarie Coal Complex Transformation Precinct Master Plan 2026* (abbreviated as Master Plan).

Land to which this Master Plan applies

The draft Master Plan applies to the Macquarie Coal Complex Precinct (the Precinct) as shown in the aerial below (Figure 1). Land subject to this draft Master Plan is owned by Macquarie Coal Joint Venture.

Post mining land use policy context

The Macquarie Coal Complex Pilot Project was initiated as part of the NSW Government's wider response to the *Beneficial and productive post-mining land use Parliamentary Inquiry* (2025). The Inquiry recommended a comprehensive review of the post mining regulatory framework to support

innovative, productive and employment-generating reuse of mined land, including provision for multiple successive land uses and the modification of existing consents.

In July 2025, the NSW Government committed to undertake two pilot projects to demonstrate how alternative post mining land uses can be delivered under the existing planning and regulatory framework. The Macquarie Coal Complex is one of these pilots and the rezoning was progressed by the Department of Planning, Housing and Infrastructure (the Department) in partnership with Lake Macquarie City Council (Council), Australian Government through Net Zero Economy Authority (NZE), Glencore and other NSW Government agencies.

Post mining land such as the Precinct represents a significant opportunity to deliver new employment, investment and economic diversification outcomes for the Hunter and for NSW.

Figure 1: Aerial view indicating the area subject to the draft Master Plan

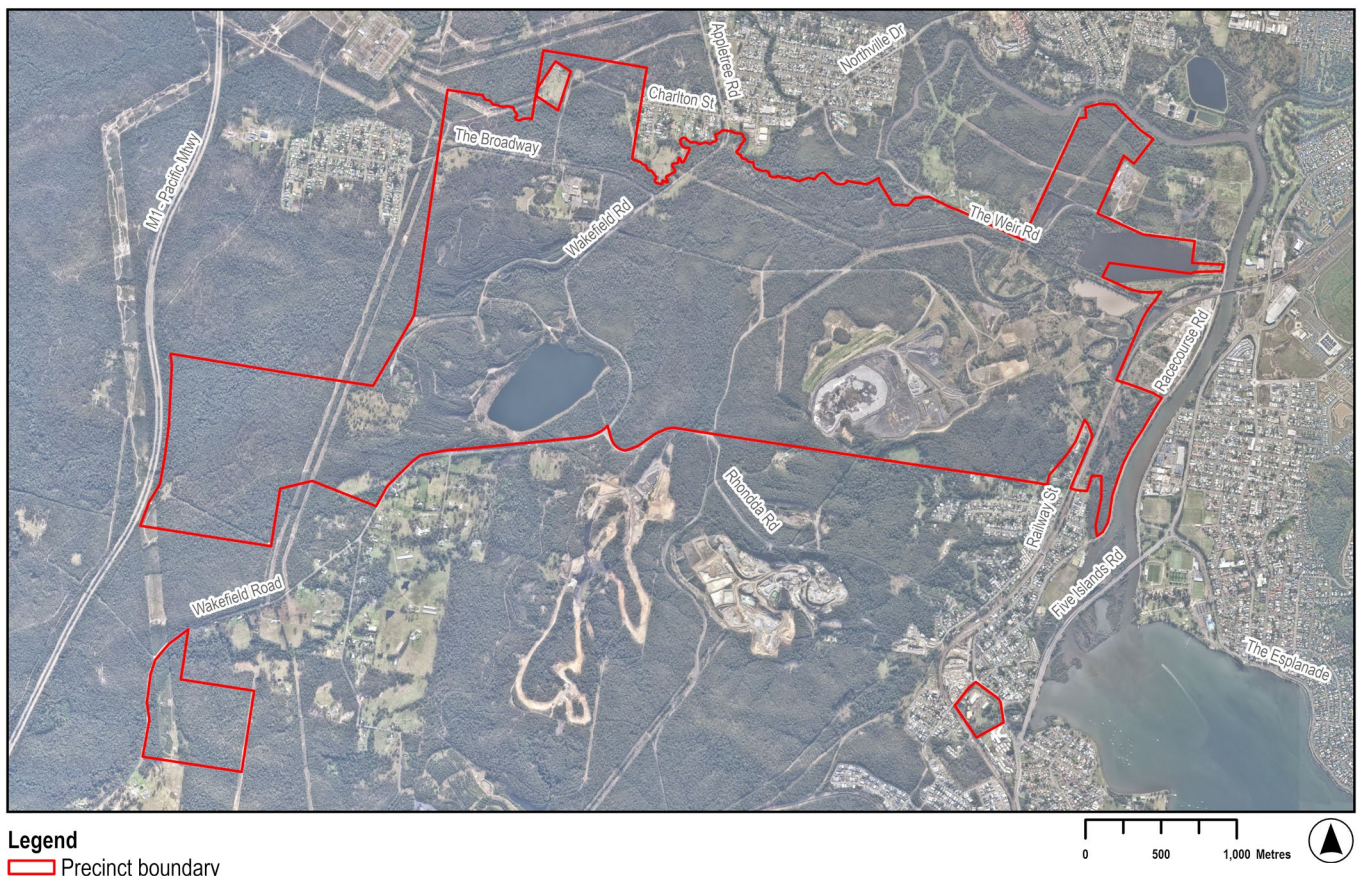


Figure 2: Process of the Precinct transforming to an employment generating precinct



Consistent with this policy direction, the Department and Council have undertaken master planning informed by technical studies to support the coordinated transformation of the former mine site to alternative, employment-generating land uses in the Precinct (Figure 2). The draft Master Plan establishes the design framework to realise this opportunity, providing the certainty needed by investors, industry partners and the community, demonstrating a transferable model for productive post mining land use across NSW.

Relationship to other planning instruments

State Environmental Planning Policy (Precincts – Regional) 2021

A new chapter will be prepared for the Macquarie Coal Complex Precinct in the *State Environmental Planning Policy (Precincts – Regional) 2021* (Precincts – Regional SEPP). The chapter will:

- Provide zoning and land use controls
- Determine permissible and prohibited land uses
- Set general and site-specific provisions
- Present the role and requirements of the draft Master Plan
- Excise pilot project sites from relevant Local Environmental Plans.

Existing planning pathways will be utilised.

Macquarie Coal Complex Post-Mining Land Use Master Plan

Prepared by the Department and approved by the Minister of Planning and Public Spaces, the draft Master Plan is a statutory document that supports the Precincts – Regional SEPP and:

- Provides the strategic vision, guiding principles, strategic context and describes overarching Precinct opportunities and constraints
- Provides an overarching structure plan and aims and performance criteria to make sure objectives are met
- Must be considered by the consent authority when determining an application for consent
- Identifies environmental, transport and utility infrastructure to support future development within the Precinct.

The *Lake Macquarie Local Environment Plan 2014* and *Lake Macquarie Development Control Plan 2014* do not apply to the Precinct.

Figure 3: Proposed rezoning planning framework



Purpose and how to use this draft Master Plan

The purpose of this draft Master Plan is to provide a hierarchy of Precinct-wide aims and establish performance criteria to support the delivery of a diverse range of land uses that support regional economic diversification, attract investment and create local jobs and protect high value biodiversity (Figure 4). It is intended to establish a broad framework for implementation where detailed design, operational and market feasibility must be balanced to achieve the vision. The draft Master Plan is to be applied with flexibility and the consent authority may allow for alternate viable solutions so that otherwise permissible development can be carried out.

This draft Master Plan details the overall planning direction for the Precinct and is not prescriptive with regards to built form design solutions, rather sets outcomes to be achieved through flexible solutions, such as adaptive reuse and staged delivery.

Development applications must:

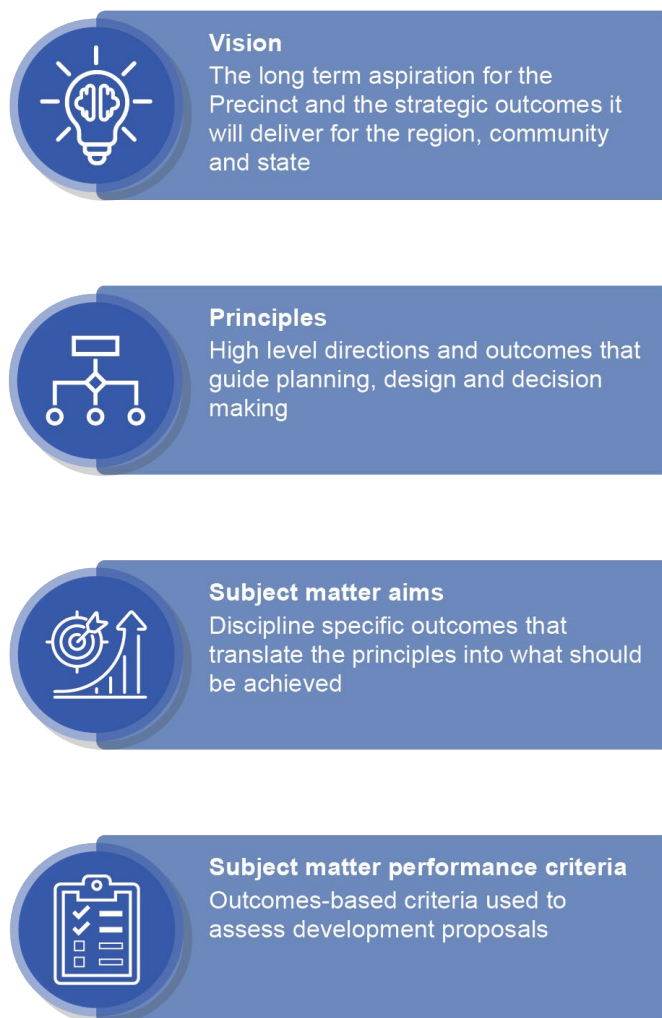
- Consider the relevant aims and performance criteria
- Provide site-specific technical assessments (such as access, flooding, drainage, water quality, tree preservation, utilities, contamination and acoustic separation) as required
- Demonstrate the development's proposed use is consistent with this draft Master Plan
- Demonstrate built form suitability for the Precinct.

Future planning considerations: Addendum to Master Plan

The Precincts - Regional SEPP, Chapter 7 requires that a Master Plan apply to the Macquarie Coal Complex Precinct and that the Master Plan is to be considered in the assessment of any development application within the Precinct.

To provide for the orderly and economic development of the Precinct, some development will require an addendum to the Master Plan to be prepared before a development application can be lodged through the Planning Portal.

Figure 4: Master Plan Precinct-wide planning hierarchy



When is an addendum required?

You must prepare an addendum to the Master Plan before lodging a development application if:

- The proposal is located on land identified as environmentally constrained on the Structure Plan (Figure 9).
- The proposal is located on land identified as requiring further investigation on the Structure Plan (Figure 9).

What should the addendum include?

- A map showing proposed land uses within the sub-precinct(s).

- Guidance on what technical reports (e.g., traffic, noise, heritage, environmental) are needed at the development application (DA) stage for permissible land uses.
- Clear performance criteria and development controls for built form, amenity, and environment.
- Identification of sensitive land and hazards (e.g., flood risk, biodiversity, geotechnical, contamination).
- Buffers and controls to manage land use conflicts, especially between industrial, sensitive uses and environmentally constrained land.
- Information on infrastructure, public spaces, and how these will be delivered.
- Urban design principles, including Connecting with Country.
- Access and mobility strategy, including freight and transport upgrades.
- Water management strategy (efficiency, stormwater, sustainability).
- Any other matters the Planning Secretary requires.
- There can be one addendum for the whole Precinct or several for different stages.
- The addendum must be consistent with Chapter 7 of the Precincts - Regional SEPP.
- The proponent prepares the addendum; the Planning Secretary approves it after notifying stakeholders. Key stakeholders consulted may include Lake Macquarie City Council, NSW Environmental Protection Authority, Department of Climate Change, Energy, the Environment and Water, and Transport for NSW.

The addendum must be in place before you lodge a development application for relevant projects.

Monitoring and adaptation

Every five years, the Department will review the draft Master Plan, to:

- Assess the effectiveness of the draft Master Plan provisions in achieving its aims
- Identify the need for changes to the draft Master Plan provisions to better achieve its purpose, aims and performance criteria
- Confirm the draft Master Plan provisions remain relevant to the types of development occurring within the Precinct.

At the end of each five-year period, following the completion of the first review of this draft Master Plan, consultation with relevant stakeholders including Lake Macquarie City Council will be required to be undertaken. During this review, submissions are to be considered by the Planning Secretary in the finalisation of the amended draft Master Plan.

Key points

- The addendum bridges the gap between the high-level draft Master Plan and detailed development applications.
- It provides site-specific detail without duplicating the draft Master Plan.
- It makes sure large or complex proposals are assessed within a clear, precinct-specific framework.

2 Strategic context

Precinct location and strategic fit

The Precinct comprises approximately 1160 hectares within the Lake Macquarie Local Government Area on the southern periphery of greater Newcastle. Macquarie Coal Complex is a former mining operation comprising West Wallsend Colliery, Teralba Colliery, Westside Mine, the Macquarie Coal Preparation Plant (MCP), Tailings Storage Facility (TSF) and associated infrastructure.

Coal mining has been a defining land use in the area since the early 19th Century, shaping both the physical landscape and the economic profile of surrounding communities. The mining operations in the Precinct are currently closed and undergoing rehabilitation, presenting a large, consolidated landholding with significant potential for new employment-generating uses.

The Precinct is characterised by a diverse post mining landscape that includes rehabilitating landforms, residual industrial infrastructure, water management features and areas of ecological regeneration. These attributes create a complex but opportunity-rich setting. Extensive brownfield areas with existing access and servicing potential are complemented by areas of environmental value requiring protection and integration into future land use outcomes.

Transforming the Macquarie Coal Complex aligns with the strategic directions of the *Hunter Regional Plan 2041*, which seeks to diversify the regional economy, leverage existing infrastructure and repurpose former mining land to support long-term employment growth. At a local level, the Precinct supports the *Lake Macquarie Local Strategic Planning Statement 2025–2045*, particularly in relation to growing and diversifying the economy, protecting environmental assets and delivering infrastructure that supports productivity and sustainable growth.

Figure 5 illustrates the Precinct's location and regional context.

Regionally significant employment opportunity

The Hunter region is undergoing a structural economic diversification. Driven by decarbonisation and shifting global energy markets, the region is forecast to lose around 14,000 coal-related industrial jobs by 2035 (Aurecon, 2026). Total employment is expected to grow by 40,000 to 75,000 jobs over the next decade through expansion in healthcare, education, professional services and emerging clean economy sectors (Aurecon, 2026). The Hunter's established energy and industrial infrastructure, strategic port and freight networks, skilled workforce and research institutions position it strongly to capture this growth. Realising this potential depends on supporting changing workforce, delivering strategic infrastructure and providing well-located investment-ready employment land.

Lake Macquarie is at the centre of this transformation. With an aging population that is forecast to grow by 26% to 2046, together with high regional immigration, the city is under sustained pressure to deliver new jobs and services. Population growth is currently not matched by the supply of well-located employment land, generating a shortfall in serviced employment land by 2046 (Aurecon, 2026). At

the same time, the Lower Hunter currently lacks purpose-built, large-scale industrial precincts capable of accommodating renewable energy, advanced manufacturing, circular economy models and regional logistics uses.

The Precinct's location, scale and infrastructure position it to deliver new employment land, support workforce transformation and attract emerging industries in the Hunter.

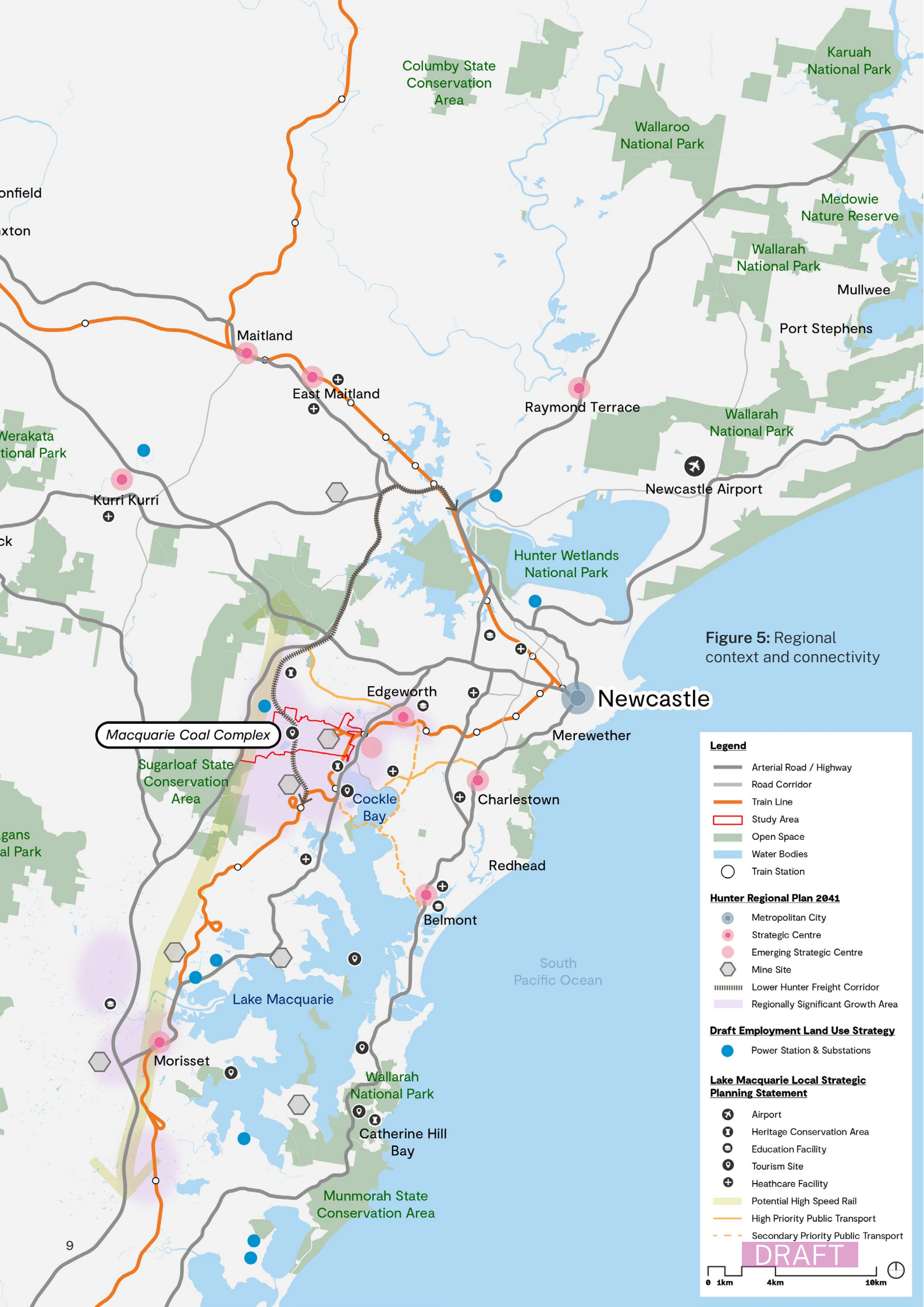


Figure 5: Regional context and connectivity

Legend

- Arterial Road / Highway
- Road Corridor
- Train Line
- Study Area
- Open Space
- Water Bodies
- Train Station

Hunter Regional Plan 2041

- Metropolitan City
- Strategic Centre
- Emerging Strategic Centre
- Mine Site
- Lower Hunter Freight Corridor
- Regionally Significant Growth Area

Draft Employment Land Use Strategy

- Power Station & Substations

Lake Macquarie Local Strategic Planning Statement

- Airport
- Heritage Conservation Area
- Education Facility
- Tourism Site
- Healthcare Facility
- Potential High Speed Rail
- High Priority Public Transport
- Secondary Priority Public Transport

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0 1km 4km 10km

Precinct opportunities



A generational-scale opportunity to transform former mining land into the Lower Hunter's premier employment precinct, delivering diverse jobs and much-needed employment land alongside lasting community and environmental outcomes.



Unmatched scale under single ownership – one of the largest master planned employment land releases under single ownership in NSW, providing flexible lot configurations, co-location of complementary industries and the ability to accommodate large-footprint uses.



A skilled workforce – one of Australia's deepest pools of transferable industrial skills in mining, heavy industry, engineering, logistics, construction and environmental management, directly applicable to the advanced manufacturing, circular economy models, energy-intensive, technology and logistics uses targeted for the Precinct.



Strategic multi-modal connectivity – direct access to motorway, arterial, rail, port and airport networks supports a diverse range of employment uses over the long term.



Development-ready brownfield land – extensive previously cleared and disturbed land, established internal access roads, hardstand, servicing corridors and modified landforms reduce lead times and upfront costs compared with greenfield alternatives.



Existing energy and utility infrastructure – proximity to electricity transmission and utility infrastructure provides a foundation for diverse employment uses, including energy-intensive industries that are increasingly constrained by grid access elsewhere.



Integrated environmental, cultural and employment outcomes – rehabilitating landforms, native bushland, waterways and ecological corridors provide the framework for a distinctive precinct identity that integrates diverse employment opportunities with high-quality landscape, biodiversity, recreation and Connecting with Country outcomes.

Precinct constraints



The Precinct presents a range of environmental, physical and infrastructure constraints that are managed through the structure plan and addressed progressively through staged development and site-specific assessment at the development application stage.



Geotechnical stability and risk management – Development on the TSF and the land located in the dam break inundation zone to the north is currently limited. Soft alluvial soils located around the coastal wetlands in the north of the site influence construction methods.



Environmental – Existing areas of high biodiversity values, high bush fire risk and flood-affected areas require further investigation to inform development siting and design.



Sensitive uses – Interfaces between existing land uses, including residential receivers and potential future industrial land uses will need to be managed to mitigate negative noise, odour and visual impacts.



Infrastructure – Heavy vehicle access to parts of the site is currently limited due to steep gradients, access through local residential areas and existing rail crossings. The location of the existing oil pipeline, electrical transmission lines and their associated easements, setbacks and design measures will need to be considered.

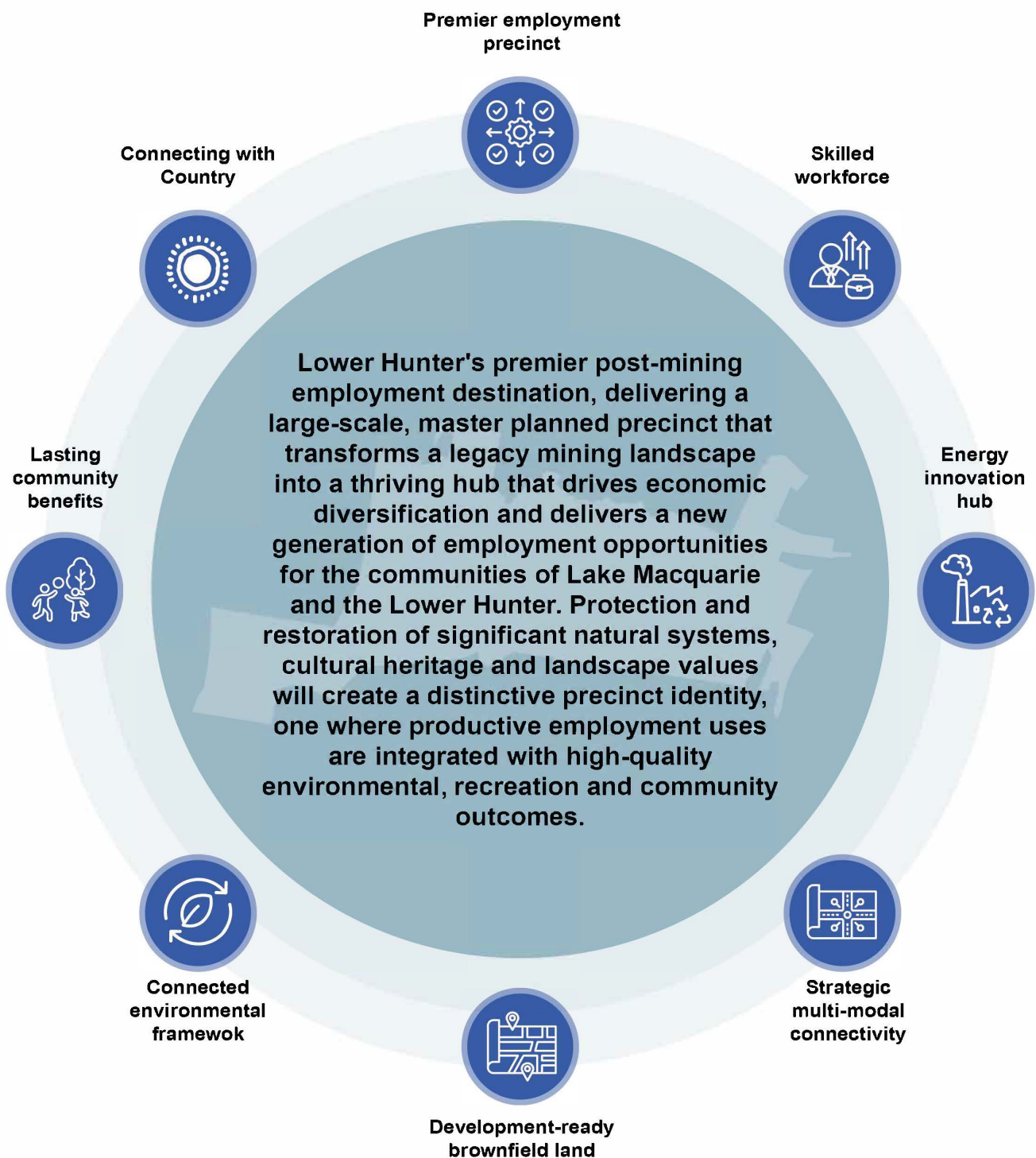


Utility infrastructure – Viability and sequencing of the provision of utilities, particularly to the West Lake sub-precinct, will need further investigation.

3 Macquarie Coal Complex Precinct

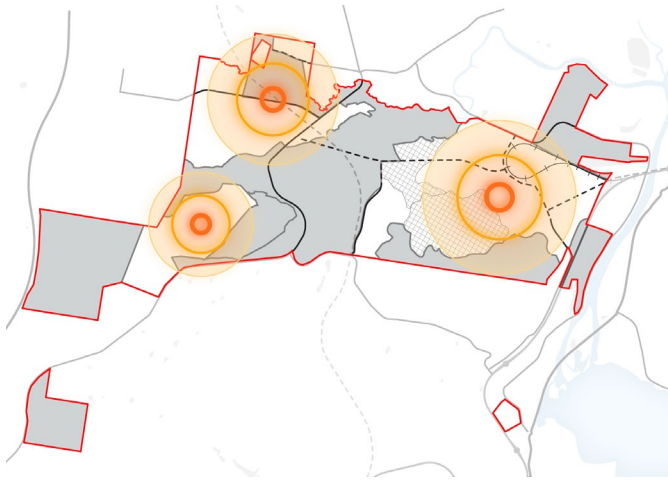
Precinct vision

Figure 6: Vision: A new generational-scale employment precinct for the Hunter



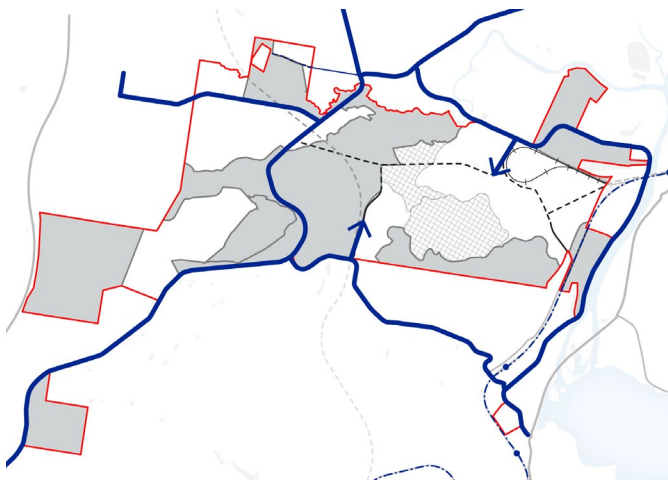
Guiding principles

The guiding principles establish the Precinct-wide directions for land use, infrastructure, environmental management, design quality and community benefit. They are to guide future planning, development assessment, addenda to the draft Master Plan and staged implementation.



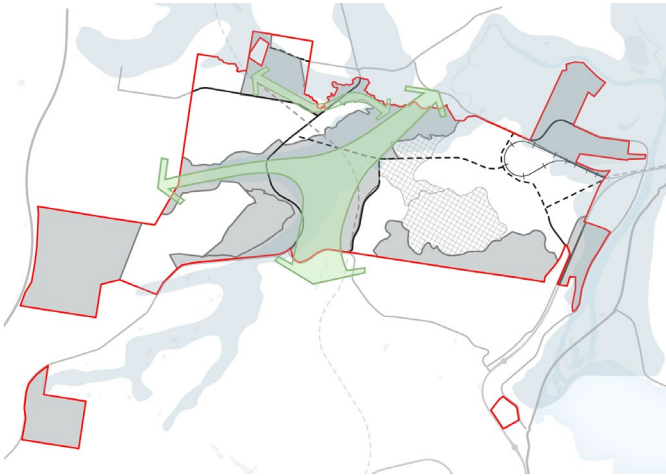
Economic diversification and employment

- A generational-scale opportunity to transform former mining land into the Lower Hunter's premier employment precinct, delivering diverse jobs and well-located employment land at a regionally significant scale.
- Enable a diversity of employment-generating land uses that support regional economic diversification, attract investment and create local jobs.
- A master planned, single-ownership Precinct that attracts investment through flexible lot configurations, co-location of complementary industries and accommodation of large-footprint uses.
- A flexible, staged delivery framework that aligns with mine closure commitments, infrastructure capacity and changing economic, social and market conditions, for the orderly transformation from rehabilitation to productive post mining uses.



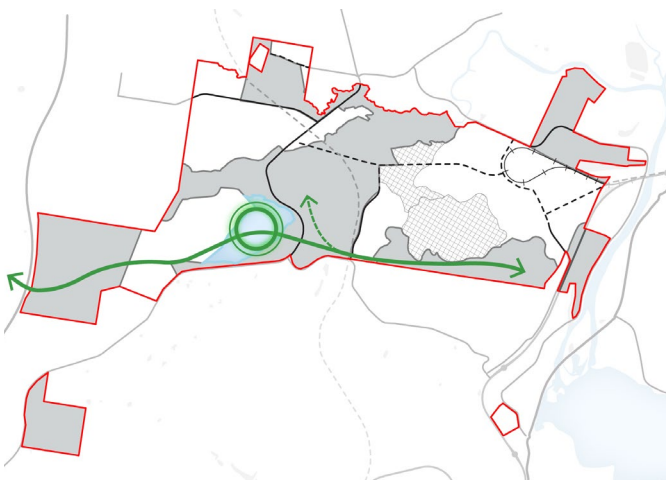
Infrastructure and mobility

- Leveraging existing transport connections to deliver an efficient, well-serviced employment precinct, with a legible internal road hierarchy that supports safe and equitable access for freight, workforce, public transport and active transport users.
- Utilities and services that meet the current and future servicing needs of the Precinct, whilst minimising land impacts and maximising reliability, efficiency and sustainability.



Environment and resilience

- A connected network of high-value biodiversity, wetlands and riparian corridors retained as integral elements of the Precinct.
- Encourage restorative and resilient development that responds to biodiversity values, flood-prone areas, drainage systems and minimises cumulative impact.
- Land use and development enable a coordinated transformation to net-zero emissions.
- A precinct-scale approach to managing post mining constraints through a staged, risk-informed framework that progressively develops land as conditions mature.
- Support circular economy models through the co-location of complementary industries that can share resources, infrastructure and services, reducing waste and improving resource efficiency.



Place and community benefits

- A distinctive post mining landscape identity, where rehabilitating landforms, remnant native bushland, waterways and ecological corridors shape the Precinct's character and sense of place.
- High-quality urban design and public realm outcomes with built form, subdivision and interface treatments that respond to topography, rehabilitated landforms and the broader environmental and cultural setting.
- Integrated community-supporting facilities co-located with employment areas and public open space, delivering tangible community and public benefit at a scale that complements the Precinct's primary employment function.
- Leveraging West Lake and environmentally constrained areas to provide accessible recreation, nature-based tourism and connected trail and active transport network for workers, surrounding communities and visitors.

Connecting with Country

The Precinct is located on Awabakal Country, adjacent to the shores of Cockle Bay in the north of Lake Macquarie. The draft Master Plan recognises Country as a living system of interconnected cultural, ecological and physical relationships, including land, waters, skies, plants, animals, cultural practices, stories and community. It also recognises that planning for the transformation of a former mining landscape must be informed by an understanding of Country and respectful engagement with Aboriginal people who hold cultural knowledge, rights and responsibilities for the area.

The draft Master Plan has been prepared as an early framework for the transformation of the Precinct from mining to sustainable and productive post

mining land uses. It establishes the foundation for planning, design and development to be guided by the principles of the Government Architect NSW *Connecting with Country Framework*.

As the draft Master Plan is exhibited and refined, meaningful engagement with Lake Macquarie Aboriginal community stakeholders will form an important part of the process to refine and finalise the draft Master Plan. This should include engagement with Traditional Custodians, Aboriginal knowledge holders, the Awabakal Local Aboriginal Land Council, the Biraban Local Aboriginal Land Council and other Aboriginal organisations or representatives identified through the engagement process.

Credit: Glencore



4 Structure plan and sub-precinct framework

Structure plan

The structure plan (Figure 9) is the principal spatial framework for the draft Master Plan. It identifies the arrangement of developable areas, environmentally constrained areas, recreational areas, movement networks and key constraints across the Precinct to guide development.

The structure plan outlines the spatial logic for future development at a precinct scale, providing a framework within which detailed design, subdivision and development assessment will occur. The structure plan is supported by technical studies.

The following categories are used to describe land within the structure plan:

- **Developable areas** – Land that is potentially suitable for development, subject to further technical investigation and site-specific assessment at the development application stage.
- **Environmentally constrained areas** – Land identified as having biodiversity values or where natural features (watercourses, wetlands, steep slope, flooding) constrain development.
- **Recreation areas** – Land potentially suitable for active and passive recreation facilities, balanced with protecting biodiversity and cultural values.
- **Further investigation** – Potentially developable land that is constrained in the short to medium term by geotechnical, dam safety, biodiversity or flood risk factors, requiring resolution before development can proceed. Development of this land is contingent upon the completion of relevant technical investigations and the achievement of applicable staging triggers.

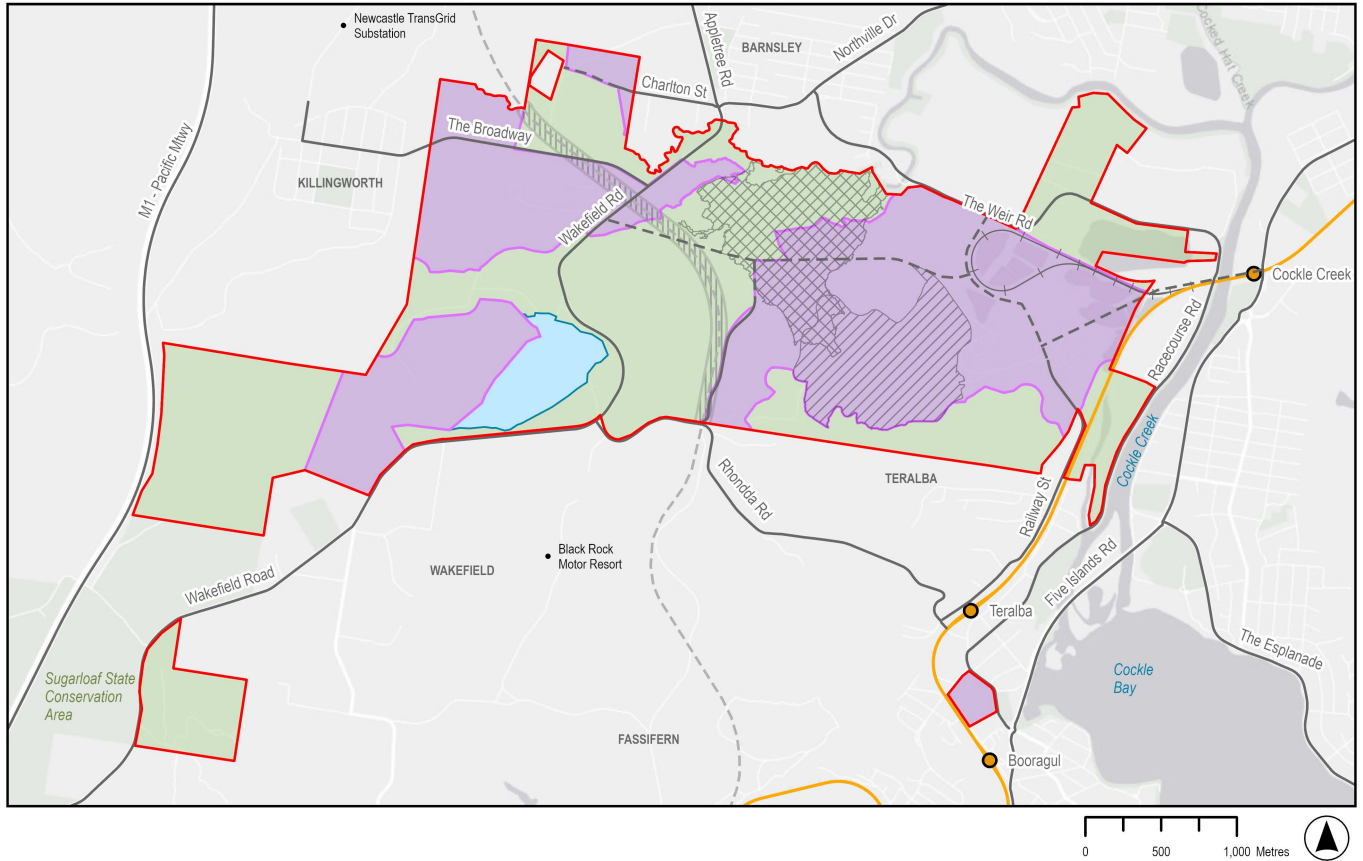
Figure 7: Indicative visualisation east over Macquarie Coal Complex Precinct



Figure 8: Indicative visualisation west over Macquarie Coal Complex Precinct



Figure 9: Macquarie Coal Complex Precinct Structure Plan

**Legend**

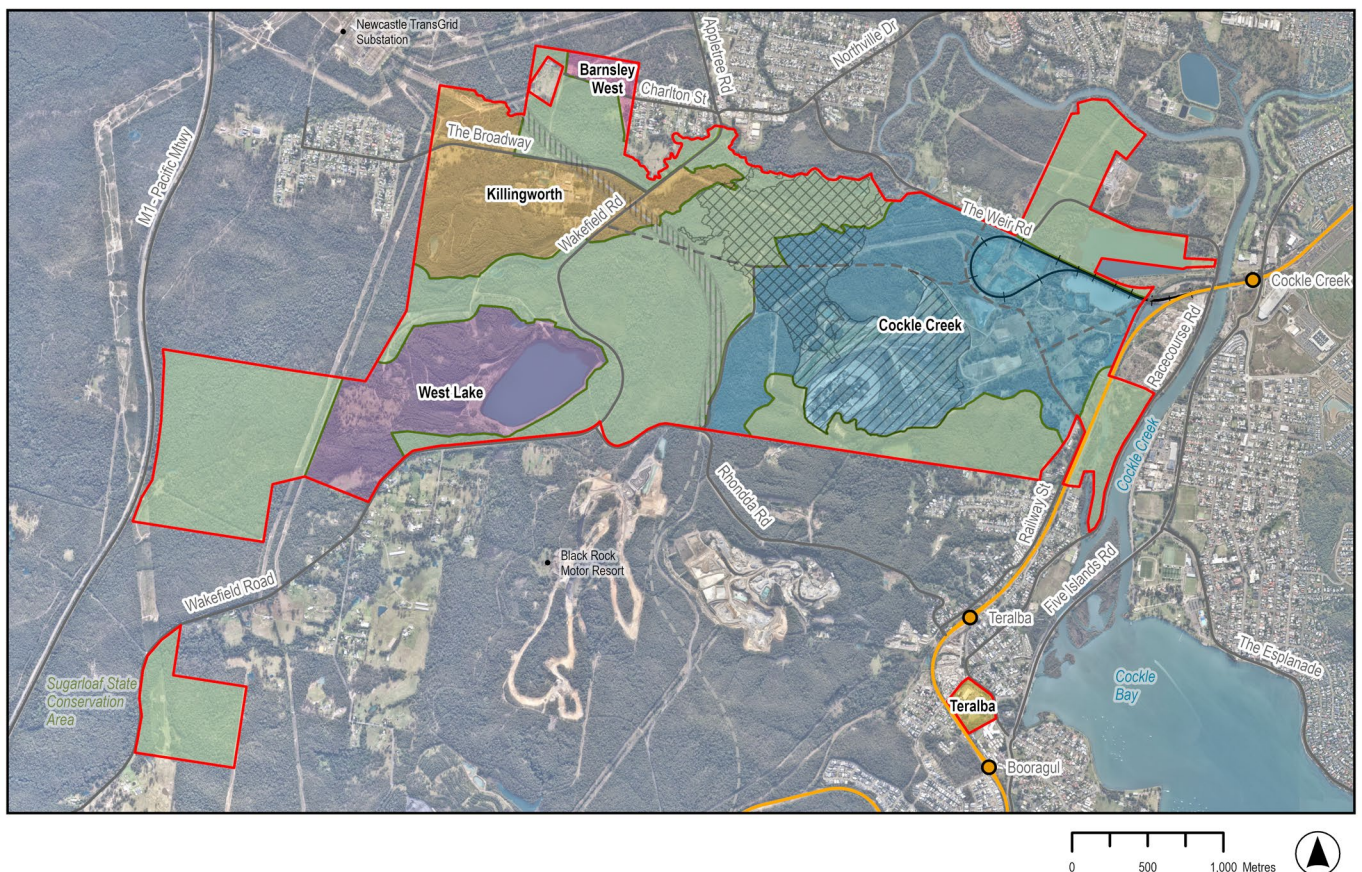
- ▬ Precinct boundary
- Railway station
- +— Rail loop
- Railway line
- Road
- - - Road - potential
- - - Indicative Lower Hunter Freight Corridor
- Environmentally constrained areas - further investigation required
- Developable areas
- Recreation areas
- Further investigation area - dam break inundation zone
- Further investigation area - tailings storage facility
- Further investigation area - Lower Hunter Freight Corridor

Sub-precinct framework

The structure plan organises the Precinct into five development sub-precincts (Figure 10), set within a connected network of environmentally constrained areas that define the Precinct's landscape framework. Each sub-precinct presents a distinctive opportunity, shaped by its topography, legacy mining features, environmental values, access and proximity to existing communities and infrastructure.

Together the sub-precincts enable a coordinated, staged transformation of the Precinct that responds to physical setting while delivering a diverse range of employment, recreation and community outcomes. Each sub-precinct description presents key characteristics, land use intent and opportunities for future planning and development, against which development proposals will be assessed.

Figure 10: Sub-precincts



Legend

— Precinct boundary

● Railway station

— Rail loop

— Railway line

— Road

--- Road - potential

--- Indicative Lower Hunter Freight Corridor

Environmentally constrained areas - further investigation required

Bamsley West (light industrial)

Cockle Creek (employment)

Killingworth (employment)

West Lake (mixed use)

Teralba (residential)

Further investigation area - dam break inundation zone

Further investigation area - tailings storage facility

Further investigation area - Lower Hunter Freight Corridor

Land use intent

Future development will be anchored by employment-generating activity that supports regional economic diversification, attracts investment and creates local jobs. Complementary commercial, recreation, community and visitor land uses will support the workforce, deliver community benefit and contribute to the broader public value of the post mining transformation.

Figure 11: Employment land use envisaged in the Precinct

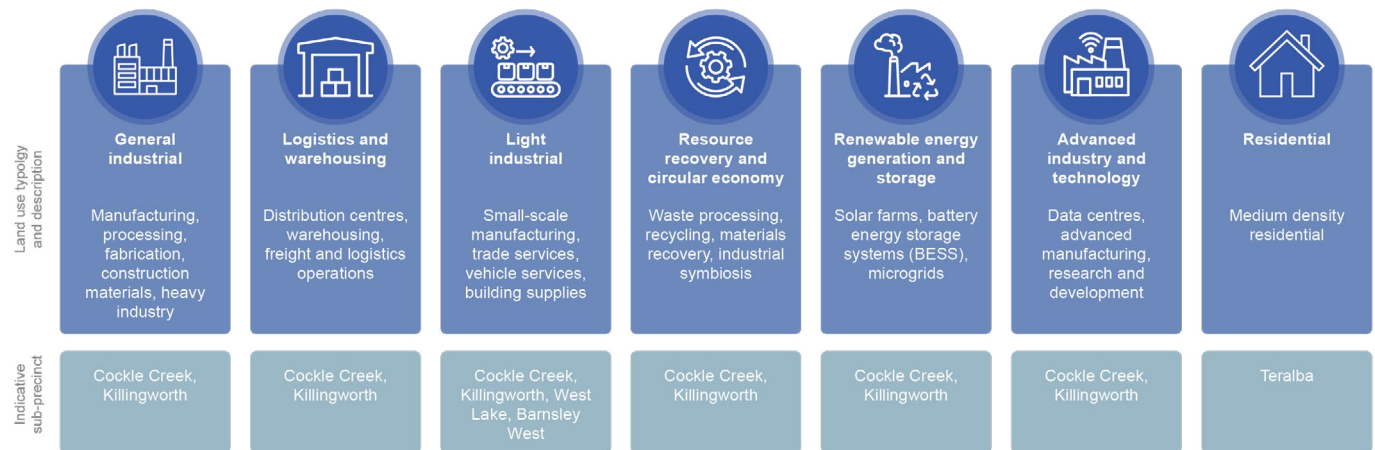
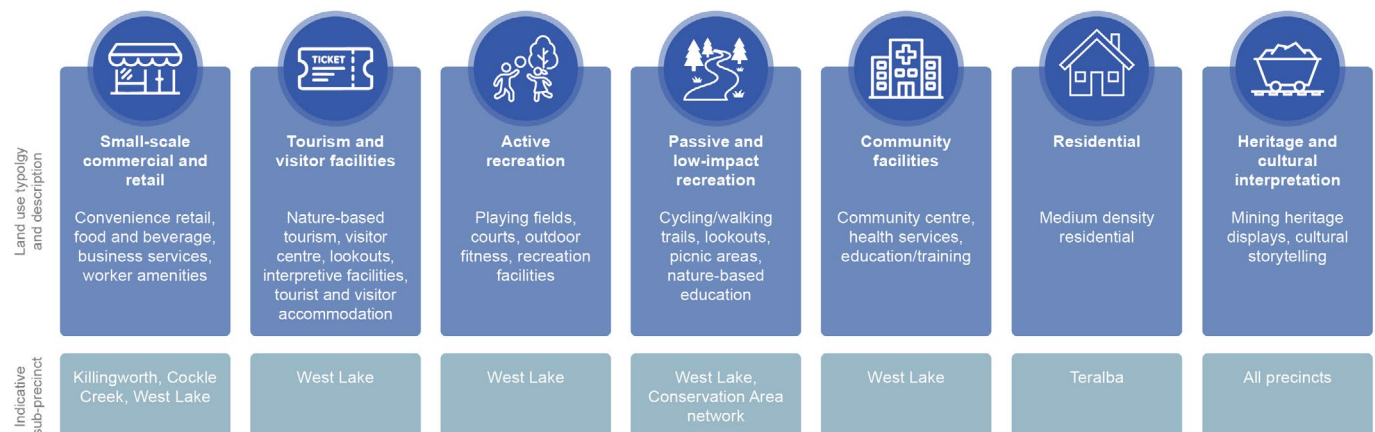


Figure 12: Complementary land use typologies envisaged



Cockle Creek

Characteristics

Cockle Creek sub-precinct is located in the eastern portion of the Precinct and encompasses the former MCPP, TSF, mine rail loop and associated operational infrastructure. Its direct interface with the Main Northern Railway, together with established access from Racecourse Road, Railway Street and The Weir Road, positions the sub-precinct as a strategically connected gateway within the broader Precinct.

Historic industrial use has shaped a substantially modified landscape of previously disturbed land and legacy infrastructure. These attributes provide a strong foundation for early activation and support the efficient reuse of land.

Key opportunities

The following attributes shape the sub-precinct's development context and inform a flexible, opportunity-led planning response:

- Reuse of legacy industrial assets – The former coal preparation plant, mine rail loop and existing rail interface with the Main Northern railway provide a platform for new industrial, manufacturing, energy and rail-related uses that leverage available infrastructure.
- Early activation of available land – Areas outside the TSF and dam break inundation zone are well-positioned for near-term development, with further land progressively released as TSF rehabilitation advances.
- Enhanced connectivity – A potential new bridge crossing of Cockle Creek is identified for investigation to strengthen road access and unlock additional development opportunities across the sub-precinct.
- Improved internal movement networks – New internal road links will support integrated movement across the sub-precinct, enabled by revocation of the TSF dam declaration or delivery of bridge infrastructure above the inundation level.
- Supporting industries that would facilitate rail manufacturing, including workshops with skilled trades.

- Established access for short term activity – Existing connections from Railway Street, Racecourse Road, The Weir Road and Rhondda Road provide immediate access to developable land beyond the dam break inundation zone, supporting early-stage uses.
- Integration of environmental values – High value biodiversity, fauna habitat and wetland areas to north and east of the sub-precinct are retained and integrated into the land use pattern, supporting a considered balance between development and environmental outcomes.

Figure 13: Indicative mixed industrial development concept



Figure 14: Indicative mixed industrial development concept utilising rail connectivity



Killingworth

Characteristics

Killingworth sub-precinct is located in the north-western portion of the Precinct, near established residential areas of Killingworth and Barnsley. Accessible topography, established road connections via Wakefield Road and The Broadway, and a direct interface with environmentally constrained areas and biodiversity corridors to the south and east define its place-based setting.

Its proximity to existing communities and the prevalence of previously disturbed land from former mining operations position the sub-precinct as a strong candidate for early activation. These attributes provide a practical platform for short-term development opportunities, aligned with rehabilitation outcomes and sequenced provision of servicing.

Key opportunities

The following attributes shape the sub-precinct's development context and inform an opportunity-led planning response:

- **Compatible community interface** – Interface with established residential areas of Killingworth and Barnsley is managed through appropriate buffers, setbacks and acoustic treatments to make sure land use compatibility and protect amenity.
- **Efficient access and movement** – Accessible topography and established connections through Wakefield Road and The Broadway enable efficient site access, with heavy vehicle routing strategy that directs freight to approved routes and reinforces the sub-precinct's logistics advantages.
- **Energy-enabled employment opportunity** – Proximity to the Transgrid substation positions Killingworth to attract energy-intensive and data-driven industries seeking reliable grid access, supporting the Precinct's role in the Hunter's energy and digital economy transformation.
- **Local jobs** – The sub-precinct's location adjoining established communities provides for accessible local employment that supports workforce participation and liveability across the surrounding area.

- **Integration of environmental values** – Interfaces with environmentally constrained areas and biodiversity corridors inform a considered land use pattern that retains ecological connectivity while enabling productive use of previously disturbed land.
- **Evidence-based sequencing** – Geotechnical, water quality, utilities and access investigations will inform the staged delivery of development.

Figure 15: Indicative light industrial development concept



Figure 16: Indicative light industrial layout development concept



West Lake

Characteristics

West Lake sub-precinct occupies the south-western portion of the Precinct and encompasses the former Westside void, now a water-filled lake, together with the surrounding rehabilitation areas. The sub-precinct's distinctive landscape identity is shaped by the interplay of water, remnant bushland, rehabilitated landforms and elevated topography, with expansive views across the Precinct and towards Lake Macquarie.

The lake is a signature landscape asset that anchors a unique sense of place. It provides the foundation for nature-based experiences, recreation and visitor infrastructure that enhance the Precinct's broader attractiveness as an employment and lifestyle destination. The sub-precinct enables an active waterfront approach to commercial, tourism and recreation around the foreshore, with development carefully integrated to retain and showcase the area's biodiversity and landscape values.

Further investigation areas on the eastern side of the lake offer additional opportunity for low-impact, sensitive development that responds to the desired future character of West Lake and complements its biodiversity and landscape setting.

Key opportunities

The following attributes shape the sub-precinct's development context and inform an opportunity-led, planning response:

- A landscape-led destination – West Lake will evolve as the Precinct's landscape, community and visitor heart, where rehabilitated land, water and bushland frame a curated employment, tourism, recreation and interpretation concept that supports economic diversification and long-term adaptability.
- Signature landscape setting and visitor experience – The lake and distinctive landscape identity that supports nature-based tourism, recreation and visitor infrastructure, including walking and cycling trails, lookouts, water-based facilities and heritage interpretation.
- Integrated landscape and environmental values – Rehabilitation areas, habitat corridors and interfaces with rural landscapes and environmentally constrained areas inform a considered land use pattern, with buffer zones and edge treatments that protect surrounding values and reinforce ecological connectivity across the Precinct.
- Evidence-based sequencing – Geotechnical, water quality, utilities and access investigations will inform the staged delivery of development.

Figure 17: Indicative lakeside development concept



Barnsley West

Characteristics

Barnsley West sub-precinct is in the north-western portion of the Precinct, comprising separate parcels of land associated with the former Wakefield Colliery. Accessed via Wakefield Road and Charlton Street, the sub-precinct sits within a distinctive landscape setting shaped by its interface with adjoining residential areas, environmentally constrained lands and established transmission infrastructure. High biodiversity land between Barnsley West and the Killingworth sub-precinct reinforces its discrete identity.

Key opportunities

The following attributes shape the sub-precinct's development context and inform an opportunity-led, planning response:

- Small-scale urban services and light industry – The sub-precinct's compact scale and adjacency to residential areas provide for a curated mix of small-scale urban services, light industrial and complementary employment uses that meet local needs while supporting the Precinct's broader economic diversification.
- Amenity-led interface with Barnsley village – The residential interface informs a high-quality edge response, including built form transitions, landscape buffers, acoustic treatments and operational controls, that protect the amenity and character of the existing community.
- Coordinated infrastructure interface – Existing transmission easements are integrated into the land use pattern, informing layout, access and development footprints that align with infrastructure outcomes and unlock practical development areas.

Figure 18: Indicative small unit industrial development concept



Teralba

Characteristics

The Teralba sub-precinct is located approximately two kilometres south-east of the main Precinct, adjoining the established Teralba local centre. The sub-precinct is defined by its direct proximity to two rail stations, its interface with the heritage-listed Teralba Conservation Area, and its integration with the established residential and commercial fabric of the Teralba village.

Together, these attributes position Teralba as the Precinct's mixed use sub-precinct, a compact, well-serviced setting where new housing and complementary uses can support the ongoing vitality of the local centre and contribute to the broader regional transformation envisaged by the Master Plan.

Key opportunities

The following attributes shape the sub-precinct's development context and inform an opportunity-led planning response:

- Conveniently connected and serviced – Positioned approximately 300 metres from Booragul Station and 400 metres from Teralba Station, the sub-precinct enables direct access to the regional passenger rail network and supports a walkable, low-car neighbourhood model.
- Reinforcement of the Teralba local centre – Proximity to established retail, community and commercial services provides the foundation for residential and complementary uses that strengthen the centre's economic base, activate the public realm and support day-to-day amenity for residents and workers.
- Heritage-informed place-making – The setting of the Teralba Conservation Area and local heritage items, including the Miner's Cottages and Teralba Public School, informs a considered design response that respects established character, scale and views, and integrates heritage interpretation as a defining element of local identity.
- Landscape-led response to Cockle Creek – The Cockle Creek corridor informs a considered land use pattern that balances development opportunity with waterway health, riparian amenity and flood conveyance, delivering a high-quality blue-green edge to the sub-precinct.
- Enhanced pedestrian and cycling connectivity – New and upgraded active transport links to Teralba Station and across the railway strengthen the sub-precinct's transit-oriented potential and improve everyday connections between the neighbourhood, the local centre and surrounding communities.
- Early activation potential – Progress in mine closure and rehabilitation, together with established servicing connections, proximity to existing urban areas and direct public transport access, positions the sub-precinct as a strong candidate for early activation and near-term delivery.

Environmentally constrained areas

Characteristics

Environmentally constrained areas form an integral and defining element of the Precinct's structure, comprising tracts of remnant native vegetation, rehabilitation areas, riparian corridors, waterways and ecological networks distributed across the Precinct. Together, these areas establish a connected environmental framework that supports biodiversity, water cycle management, landscape identity and Connecting with Country outcomes.

Environmentally constrained areas anchor the structure plan in a high-quality landscape setting and reinforce ecological connectivity across the region. The intent of these lands is to protect and enhance environmental, hydrological, cultural and landscape values over the long-term, while enabling sensitive land use where suitable.

Key opportunities

The following attributes shape the sub-precinct's development context and inform an opportunity-led, planning response:

- Anchoring the Precinct's ecological framework – High biodiversity areas, key fish habitat and terrestrial corridors provide north-south and east-west ecological connectivity across the Precinct. Biodiversity corridors, coastal wetlands, mapped waterways and areas subject to Voluntary Conservation Agreements informing a clear protection framework aligned with statutory requirements.
- Considered, evidence-based approach to compatible use – Where supported by detailed biodiversity assessment, low-impact, compatible uses may be accommodated within these lands, guided by the recognised hierarchy of avoid first, minimise where avoidance is not practicable, and offset residual impacts. This approach makes sure that future use is sensitively located, limited in scale, and consistent with the retention and enhancement of biodiversity, hydrological and landscape values.
- Water cycle and landscape performance – Riparian corridors along mapped waterways

provide for flood conveyance, stormwater treatment and waterway health, contributing to a resilient water cycle framework that supports development outcomes elsewhere in the Precinct.

- Compatible low-impact recreation and visitor experience – Walking trails, lookouts, nature-based education and interpretation position the environmentally constrained areas as part of a layered land use framework that complements community benefit and economic diversification across the Precinct.

Figure 19: Indicative view of environmentally constrained areas



Figure 20: Indicative trails within the environmentally constrained areas



Implementation and sequencing

Implementation

The implementation and delivery of the entire Precinct vision is intended to be sequenced over several decades, reflecting the need to manage mine closure and rehabilitation responsibilities and to allow the future land uses to evolve in alignment with development uptake. Delivery of supporting infrastructure is intended to be staged to correlate with the development as it is implemented over time.

Development sequencing provides for infrastructure delivery that is proactive but demand led and enables early activation while supporting long-term, efficient and resilient precinct development. Infrastructure delivery is to be adaptive, demand driven and coordinated, ensuring that development does not outpace enabling infrastructure while maintaining flexibility for market led uptake. Development is to be staged in accordance with available infrastructure capacity, with priority given to feasibility of connecting with existing utilities and transport networks. Infrastructure delivery should adopt an integrated and coordinated approach across utilities and transport to minimise disruption and duplication.

Figure 21: Staging and key levers



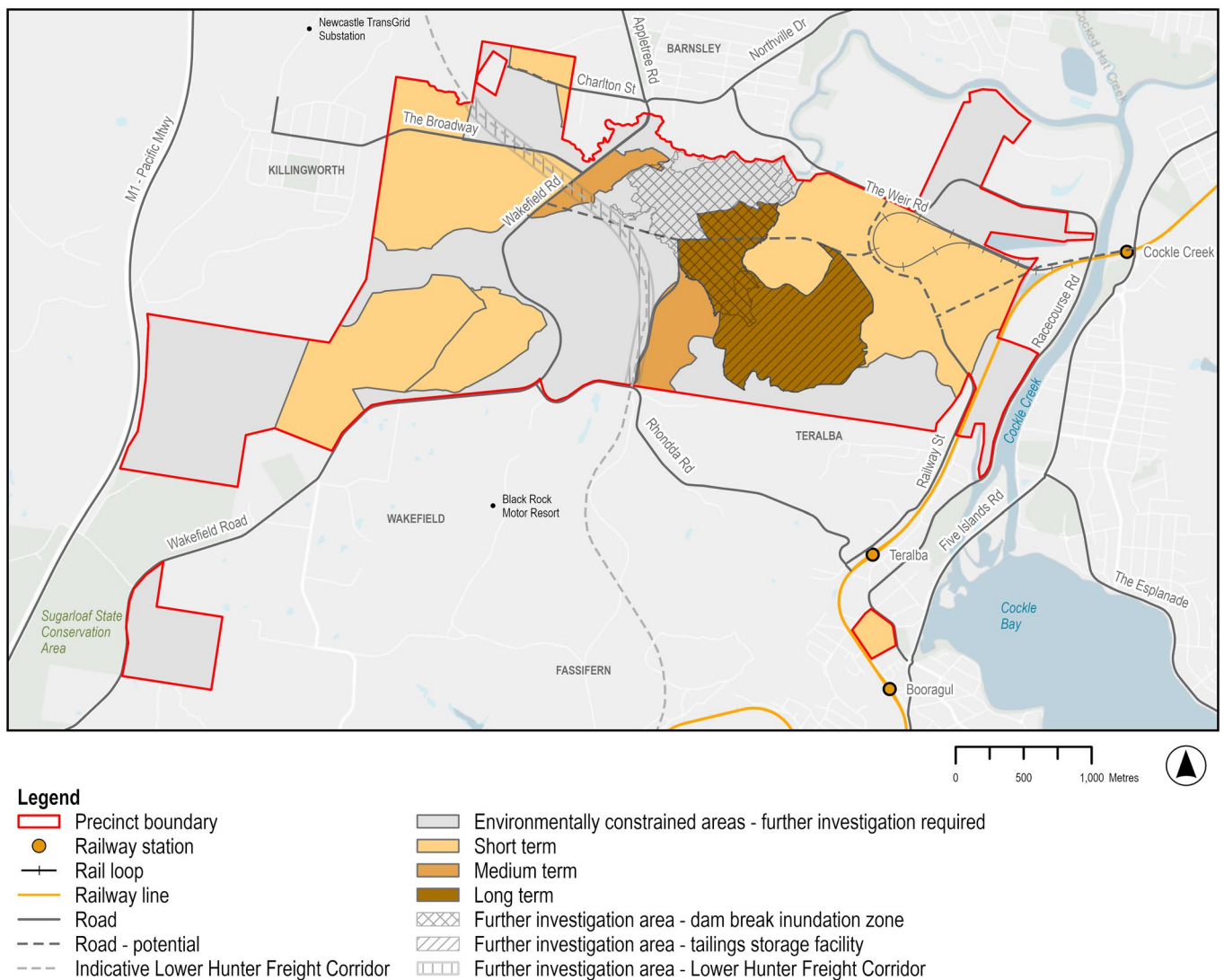
Development horizons

Development horizons are described in Table 1. Figure 22 illustrates the sequencing of development within the Precinct, which reflects the horizon within which land would commence development, subject to the drivers and staging triggers.

Table 1: Sub-precinct staging

Horizon	Timeframe	Key sub-precincts and drivers
Short-term activation	0–5 years	Cockle Creek, Killingworth, West Lake, Barnsley West, Teralba.
Medium-term activation	5–15 years	Cockle Creek, Killingworth.
Long-term activation	15+ years	Cockle Creek and Killingworth residual land.

Figure 22: Sub-precinct staging



Short-term activation

Areas identified as being capable of early release for post mining land use development share common characteristics, including land where rehabilitation is at or near completion, existing road access is available, servicing connections are feasible and the risk profile is manageable without requiring major investigations or infrastructure upgrades.

Early phases are expected to leverage available capacity within the Argenton Substation, existing potable water connections and established regional freight corridors, minimising the need for major upfront augmentation. These areas also align with transport accessibility and heavy vehicle routing capability, supporting early industrial and logistics uses.

These short-term activation areas provide the opportunity to demonstrate viability of post mining land use shift, generate investment confidence and deliver tangible employment and community outcomes.

Cockle Creek

Previously disturbed areas outside the TSF and dam break inundation zone with existing access to the regional road network. Early activation within this sub-precinct could leverage the existing rail loop with opportunities for related manufacturing facilities.

This sub-precinct benefits from the strategic role of freight and logistics in this area, leveraging proximity to supporting flexible, freight-adjacent land uses as infrastructure staging evolves.

Development in this sub-precinct is staged to align with infrastructure delivery. A graduated approach to the staging model is proposed, transitioning from lighter industry near Cockle Creek to heavier industrial and warehousing uses further within the sub-precinct.

Early activation is subject to geotechnical investigation and confirmation that development will not compromise TSF rehabilitation or increase societal risk within the dam break inundation zone.

Killingworth

The northwest area near Killingworth presents

a strategic opportunity for land uses such as advanced manufacturing, data-driven uses and business parks, which take advantage of the Transgrid substation and existing electricity capacity. The area offers opportunity for local uplift and diversification through enabling light infrastructure, business parks, technology and data-driven employment to support local jobs and complement existing residential areas.

The sub-precinct is well-suited for activation given its previously disturbed land provides a strong foundation for redevelopment, complemented by its proximity to existing high-voltage transmission infrastructure and the Transgrid substation. Additionally, road access is available via the existing local road network, with potential for future upgrades to support new land uses.

Water and wastewater servicing requires further investigation. Bush fire and biodiversity interface management will be required along the western edge.

Barnsley West

Development of previously undisturbed areas adjacent to existing residential in Barnsley is contingent upon assessment of transmission easements, servicing viability and biodiversity values.

West Lake

The West Lake area offers excellent water quality suitable for recreational use (subject to ongoing monitoring). Rehabilitation around the lake has been progressively signed off, with approximately one-third approved and the balance targeted for completion in 2026, indicating suitability for recreational contact.

Existing walking tracks and connection opportunities to Sugarloaf State Conservation Area further enhance the Precinct's potential for tourism and recreation uses that are compatible with the ecological and landscape values of the area. Basic servicing for visitor amenities would be required, while the western edge of the lake area presents ecological constraints including biodiversity corridors and remnant vegetation that define the extent of any development boundaries. Future development around the lake must maintain water quality through appropriate catchment management measures.

Teralba

Teralba sub-precinct is strategically located between the Teralba and Booragul train stations. Proximity to existing utility networks reduces servicing lead times and upfront infrastructure costs. Established road access and adjacency to nearby rail stations, residential communities and employment areas provide a ready platform for early development. Together, these attributes present a compelling opportunity to deliver residential outcomes that complement the surrounding context and contribute to the Precinct's early activation and investment momentum.

Medium-term activation

Medium-term activation areas require the delivery of enabling infrastructure, resolution of key transport and access constraints and progressive risk reduction through ongoing rehabilitation and technical investigations. These areas represent the core employment-generating lands of the Precinct and will require coordinated investment in roads, utilities and potentially rail infrastructure.

Progressive development toward the southern and more constrained areas will require coordinated upgrades to transport, utilities and Precinct scale water cycle management. This includes delivery of wastewater infrastructure (currently absent across much of the Precinct), expansion of water networks and potential major electricity augmentation or new substations where demand thresholds are exceeded. Concurrently, transport upgrades, including intersection capacity improvements, new internal road links and potential additional crossings of Cockle Creek, must be staged alongside development to maintain network performance and resilience.

Cockle Creek and Killingworth residual land

Previously undisturbed areas outside the TSF and dam break inundation zone with existing access to the regional road network. Medium term activation is subject to biodiversity investigations, geotechnical investigation and flood risk assessment.

Medium term activation in Cockle Creek sub-precinct requires confirmation that development

will not compromise TSF rehabilitation or increase risk within the dam break inundation zone.

Long-term activation

Release of areas identified for long-term development potential for productive use is contingent upon the completion of critical investigations, successful risk reduction through ongoing rehabilitation and potentially major infrastructure interventions.

Cockle Creek (TSF and dam break inundation zone)

Development of the TSF area is contingent upon several key prerequisites, including the completion and regulatory acceptance of long-term dewatering and stabilisation, together with an updated dam break failure impact assessment reflecting current capped conditions.

As a declared dam, development in both the TSF and dam break inundation zone remain limited to land uses that do not increase the risk to life or undermine the structural stability of the TSF and associated structures. The downstream dam break inundation zone may be reduced in size over time as rehabilitation progresses, potentially to the point of revoking its declaration as a dam. Once the TSF dam declaration is revoked, development downstream of the dam can be achieved, subject to other constraints.

Re-purposing and use of the TSF and associated structures can occur provided detailed structural, geotechnical and hydrological investigations occur in parallel to each development application.

Until these prerequisites are met, the TSF and its associated inundation zone may support interim or 'meanwhile' uses that maintain a low population level. Such interim uses could include solar energy installations, for which international precedents on tailings dams exist, as well as biodiversity corridor functions, landscape rehabilitation and other low occupancy uses.

Development within the further investigation areas associated with the TSF and dam break inundation areas may be brought forward to medium horizon, either as a whole or in part, depending on rehabilitation and ongoing monitoring.

Staging triggers

The progressive release of developable land across the Precinct is linked to the completion of key technical investigations and the achievement of rehabilitation milestones. The following staging triggers apply as relevant across the Precinct:

- **Dam safety and Geotech** – Geotechnical monitoring and assessment of TSF, with the view to de-declaring the dam. Updated dam break and consequence assessments demonstrating a reduction in the dam break inundation zone. Refer to **Section 9 Sustainability and resilience on page 51** for detailed dam safety aims and performance criteria.
- **Contamination** – Completion of staged contamination investigations and, where required, remediation or management measures appropriate to the proposed land use, in accordance with the *Contaminated Land Management Act 1997* and *State Environmental Planning Policy (Resilience and Hazards) 2021*.
- **Geotechnical stability** – Geotechnical assessments confirming the suitability of land for the proposed development type, particularly rehabilitated landforms.
- **Flooding** – Detailed flood modelling confirming flood planning levels, developable areas and appropriate mitigation measures for development within or adjacent to flood-affected land, including climate change and dam break interaction scenarios.
- **Infrastructure and servicing** – Confirmation of utility servicing capacity (water, wastewater, electricity, telecommunications) and delivery of enabling infrastructure to support staged development. Refer to **Section 10 Infrastructure, utilities and servicing on page 58** for detailed infrastructure and servicing aims and performance criteria.
- **Biodiversity** – Completion of detailed environmental surveys to verify biodiversity values and confirm environmentally constrained area boundaries.
- **Take-up and market demand** – Progressive development is informed by demonstrated market take-up across the Precinct.

The sequencing of development is to be coordinated with the delivery of supporting infrastructure to make sure that the transport network, utilities and community facilities can accommodate incremental growth. Development staging should also be coordinated with mine closure and rehabilitation activities so that land use transformation proceeds in an orderly manner. An indicative description of the staging triggers according to sub-precincts is provided in Table 2.

Table 2: Indicative staging triggers by sub-precinct

Sub precinct	Timing horizon	Staging intent	Key enabling triggers
Cockle Creek	Short-Medium-Long	Infrastructure-led and staged development areas linked to triggers	• Completion of rehabilitation and Geotechnical verification • Adequate provision for site access (including heavy vehicle and rail interface as required) • Delivery of enabling infrastructure and servicing • Detailed flood assessment in low-lying areas • Resolution of dam safety constraints (including reduction in dam break inundation zone extent) relevant to the proposed development and site access • Demonstrated market take-up of preceding stages within the sub-precinct or across the Precinct.
Killingworth	Short-Medium	Infrastructure-led and staged development areas linked to triggers	• Completion of rehabilitation and Geotechnical verification • Detailed flood modelling and confirmation of developable areas • Detailed flood assessment in low-lying areas • Assessment of biodiversity constraints to confirm developable land • Demonstrated market take-up of preceding stages within the sub-precinct or across the Precinct.
West Lake	Short-Medium	Infrastructure-led and staged development areas linked to triggers	• Completion of rehabilitation and Geotechnical verification • Detailed flood assessment (including climate change and interaction scenarios) • Delivery of enabling infrastructure and servicing (including consideration of onsite servicing) • Assessment of biodiversity constraints to confirm developable land • Demonstrated market take-up of preceding stages within the sub-precinct or across the Precinct.
Barnsley West	Short-Medium	Early to progressive activation	• Confirmation of servicing capacity • Local access upgrades • Verification of biodiversity and transmission easement constraints • Assessment and response to local constraints and management of interface conditions with adjacent sensitive land uses
Teralba	Short	Early activation of residential development leveraging existing infrastructure, transport and adjoining centre	• Completion of rehabilitation and Geotechnical verification • Completion of Detailed Site Investigation (DSI) and, where required, Remedial Action Plan (RAP) • Confirmation of water and wastewater servicing capacity

5 Built form and landscape

Desired future character

The Precinct offers the opportunity to deliver a distinctive future character, shaped by its post mining landscape and a diverse mix of land uses. Employment-generating activity will anchor the Precinct, complemented by commercial, recreation, community and visitor uses.

The Precinct will transform a legacy mining landscape into the Lower Hunter's leading post mining employment precinct, becoming a well-connected, productive and resilient destination that is an attractive place to work, invest, recreate and visit, with high-quality urban form and integrated landscape.



The following character principles guide Precinct development.



Built form

The Precinct supports employment generating development responsive to its landscape context setting. Building orientation, scale and siting is carefully considered to integrate with the site's natural features.



Landscape

Building on existing high environmental value, the Precinct protects and enhances its natural landscape, with remnant vegetation and waterways treated as defining landscape elements that shape the Precinct's character. Biodiversity can be protected near development areas through the establishment of green corridors, landscape buffers and setbacks that respond to topography and ecological systems.



Place and identity

The Precinct offers the opportunity to create a distinctive identity that reflects its natural qualities, cultural values and industrial history.



Public realm and mobility

Well-connected to surrounding networks, the site supports safe, clear and legible movement systems that connect sub-precincts, support active transport and prioritise pedestrian amenity and landscape integration at key interfaces. Access and circulation are designed to respond to the scale of employment uses.



Sustainability and resilience

Built form and landscape design embed water sensitive urban design (WSUD), ecological restoration, efficient resource use and climate resilience, delivering a precinct that performs as a productive landscape.



Interface and transition

Development at precinct edges manages the transition to environmentally constrained areas and surrounding communities through graduated built form, generous setbacks and landscape buffers that protect amenity and maintain natural character.



Flexibility and staged activation

Built form and subdivision patterns accommodate staged delivery and evolving uses over time, ensuring that interim conditions and early activation areas contribute positively to precinct character without compromising long-term outcomes.

Urban design and public realm

Urban design is guided by shift from mining to employment-generating uses, as well as opportunities for tourism and recreation. The design approach prioritises functional, robust and adaptable built form outcomes that support a range of industrial, logistics and contemporary employment uses.

Aims

- Achieve high-quality urban design outcomes consistent with the *Better Placed* (GANSW, 2017) objectives.
- Provide a mix of contemporary, high-quality building types and sizes to support employment opportunities that evolve in line with changing economic drivers
- Built form to be of high quality, with facades that address the street and that have articulation, modulation, passive surveillance and street activation
- Make sure building siting and scale respond to landscape, context, environmental values and interface conditions.
- Strengthen precinct identity through landscape, signage, landmarks and heritage-informed placemaking.
- Provide landscape setbacks and public realm outcomes that support amenity, biodiversity and movement.
- Create an employment precinct, with well-connected buildings, commercial nodes and integrated public and active transport.

Performance criteria

- Built form, subdivision patterns, open space and interface treatments must respond to rehabilitated landforms, engineered surfaces and the post mining landscape character of the Precinct.
- Development must contribute to a coherent precinct identity by integrating mining heritage, landscape character and, where relevant, Connecting with Country outcomes into built form, landscape and material selection.
- Buildings must respond to landscape character, topography and interface conditions.
- Building scale and height must respond to the surrounding context, consider visual impact and contribute to an appropriate variation in built form character across the Precinct.
- Road setbacks must incorporate landscape treatments that strengthen precinct character and support biodiversity and amenity outcomes.
- Where existing structures are retained, adaptive reuse must demonstrate compliance with current safety and accessibility requirements.
- Public domain, landscape and street design must integrate WSUD assets to support urban cooling, biodiversity, amenity and place identity.

Figure 23: Indicative industrial built form development concept



Figure 24: Indicative public realm development concept



Sub-precinct interface consideration

The Precinct incorporates a range of contextual interfaces, including environmentally constrained areas, open space and recreation areas, and external sensitive receivers such as established residential communities. Three interface typologies guide the design of the Precinct's edges, each shaped by the values of the adjoining land and the role of the interface in protecting amenity, biodiversity and bushfire safety outcomes:

- Typology 1 – Development adjoining environmentally constrained areas (including heavily vegetated and bushfire-prone areas)
- Typology 2 – Developable area to open space and recreation areas
- Typology 3 – Precinct boundary to external context (including adjoining residential and other sensitive receivers).

Each typology is illustrated separately in Figure 25 to Figure 27. Together, these treatments establish high-quality, well-managed edges that retain the values of adjoining land while supporting the Precinct's long-term productive use.

Aims

- Establish clear, durable transitions between developable areas, environmentally constrained areas, open space and surrounding communities.
- Protect biodiversity values, ecological corridors, waterways and cultural heritage from the indirect effects of adjacent development, including noise, light, dust, stormwater runoff, weed invasion and uncontrolled access.
- Provide that interface built form responds to the scale, character and amenity of adjoining land.
- Maintain permeability and connectivity for people while managing access to environmentally constrained areas.

Performance criteria

Typology 1 - Development adjoining environmentally constrained areas

- A. Development adjoining environmentally constrained areas protects ecological values, manages edge effects, accommodates bushfire asset protection requirements and supports a

clear, maintainable boundary between urban and conservation land.

- B. A vegetated buffer planted with locally endemic species must be provided to the environmentally constrained areas, with a baseline width of 30 metres. This width may be reduced where a site-specific ecological assessment demonstrates that a narrower buffer will achieve equivalent or improved ecological outcomes, having regard to the sensitivity of the adjoining environmentally constrained area.
- C. Where a perimeter road provides the interface between developable land and environmentally constrained area, the road reserve may form part of the buffer function.
- D. The bushland side of any perimeter road must be managed through bush regeneration to retain endemic vegetation and ecological values. The development side of the road must incorporate low-maintenance bushfire asset protection zones (e.g. wide road verges, managed grassed verges) that integrate with the urban edge.
- E. Buildings, hardstand, storage and operational activities must not be located within the ecological buffer or asset protection zone.
- F. Development intensity at sensitive environmentally constrained area interfaces is to be reduced through setbacks, lower built form scale where appropriate, and landscape screening to minimise visual and ecological edge effects.
- G. Lighting adjacent to environmentally constrained areas is to be designed to minimise light spill, glare and ecological disturbance to habitat areas.
- H. Fencing at environmentally constrained area interfaces must be in the form of a maintainable barrier that controls access to the environmentally constrained area while retaining visual permeability where appropriate.
- I. Stormwater treatment measures must be located within the developable area and outside riparian buffers and environmentally constrained areas.
- J. Edge treatments must be designed to prevent ongoing weed invasion into adjacent native vegetation, including through selection of non-invasive landscape species, ongoing

maintenance arrangements and clear physical demarcation.

Typology 2 – Developable area to open space and recreation areas

- A. Development adjoining open space or recreation areas must provide a positive, legible and active interface to the public realm.
- B. Pedestrian access points and landscape treatments are to integrate seamlessly with adjoining open space.
- C. Buildings must be oriented to support Crime Prevention Through Environmental Design (CPTED) principles and provide passive surveillance of public open space, where appropriate to the proposed use.
- D. Open space at sub-precinct interfaces are to function as a connected system that links environmentally constrained areas, walking and cycling routes, significant vegetation and natural landforms.
- E. Lighting at open space interfaces must support safe pedestrian movement while applying CPTED and wildlife-friendly lighting principles where the open space adjoins or connects to environmentally constrained areas.

Typology 3 – Precinct boundary to external context

- A. Development adjoining residential areas or other sensitive receivers must provide setbacks and landscape buffers informed by detailed land use compatibility and acoustic assessment.
- B. Buildings at sensitive interfaces must be sited and designed to minimise overlooking, overshadowing and visual dominance.
- C. Acoustic, air quality and operational mitigation measures must be incorporated where required to manage amenity impacts.
- D. Heavy vehicle access, loading and servicing areas are not to be located at residential interfaces unless no feasible alternative is available and mitigation measures are implemented to minimise noise, air quality and amenity impacts.
- E. Landscape screening must use locally endemic species and contribute to amenity and biodiversity outcomes.
- F. Boundary interface treatments must respond to the characteristics of the adjoining land, including bushfire, ecological, riparian and residential contexts.

Figure 25: Indicative interface concept – Typology 1 (Environmentally constrained areas)

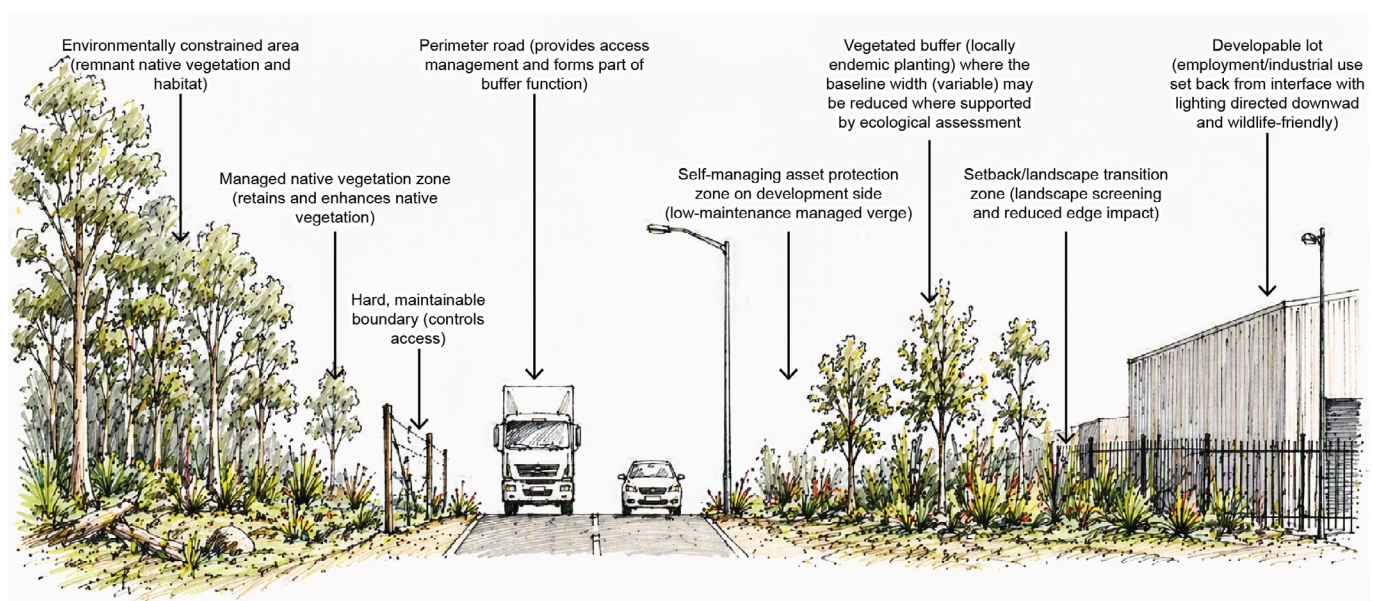


Figure 26: Indicative interface concept – Typology 2 (Open space and recreation)

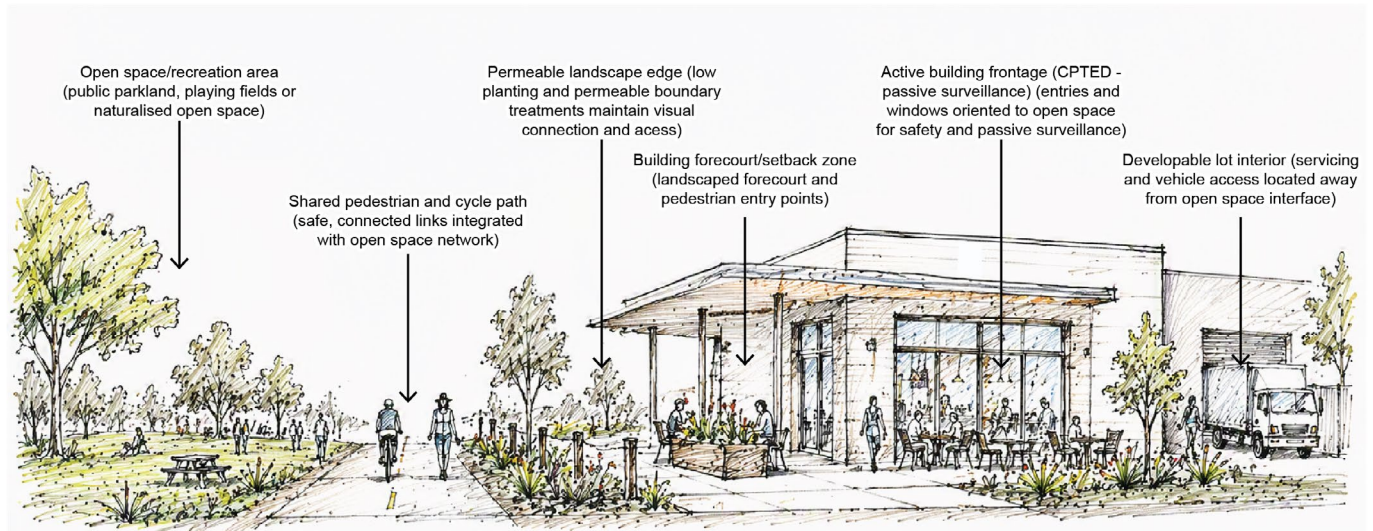
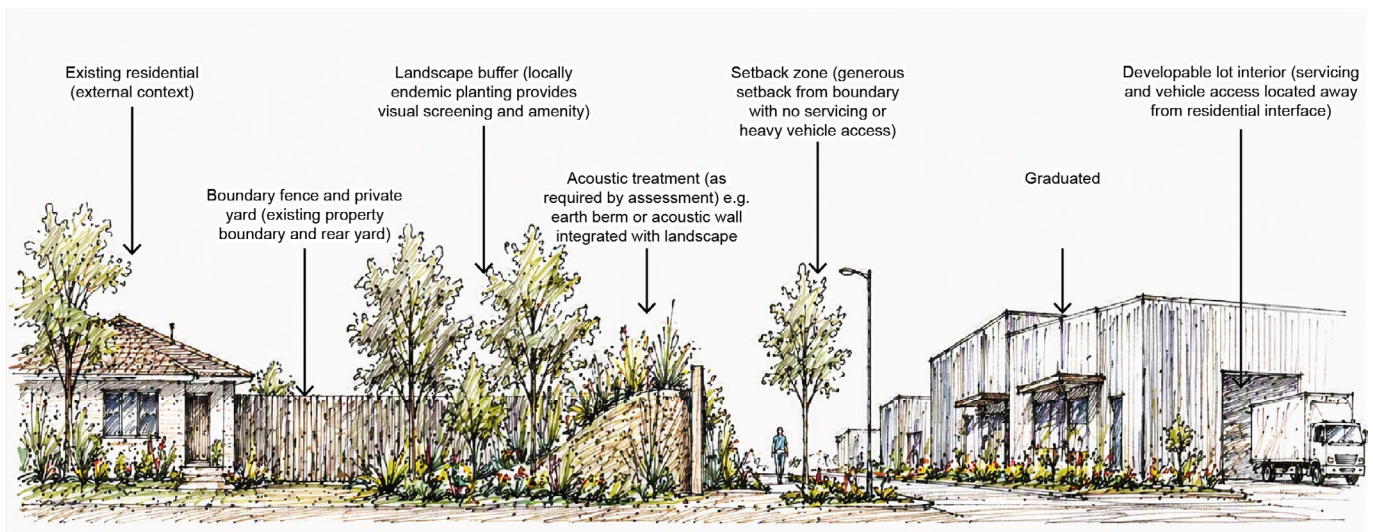


Figure 27: Indicative interface concept – Typology 3 (External context)



6 Ecology, water and landscape systems

Biodiversity and habitat corridors

The Precinct consists of a combination of previously disturbed land, rehabilitation areas and undisturbed, remnant native bushland. Vegetation includes existing cleared areas associated with the historical operation of several former mine sites, some of which have been rehabilitated back to native bushland.

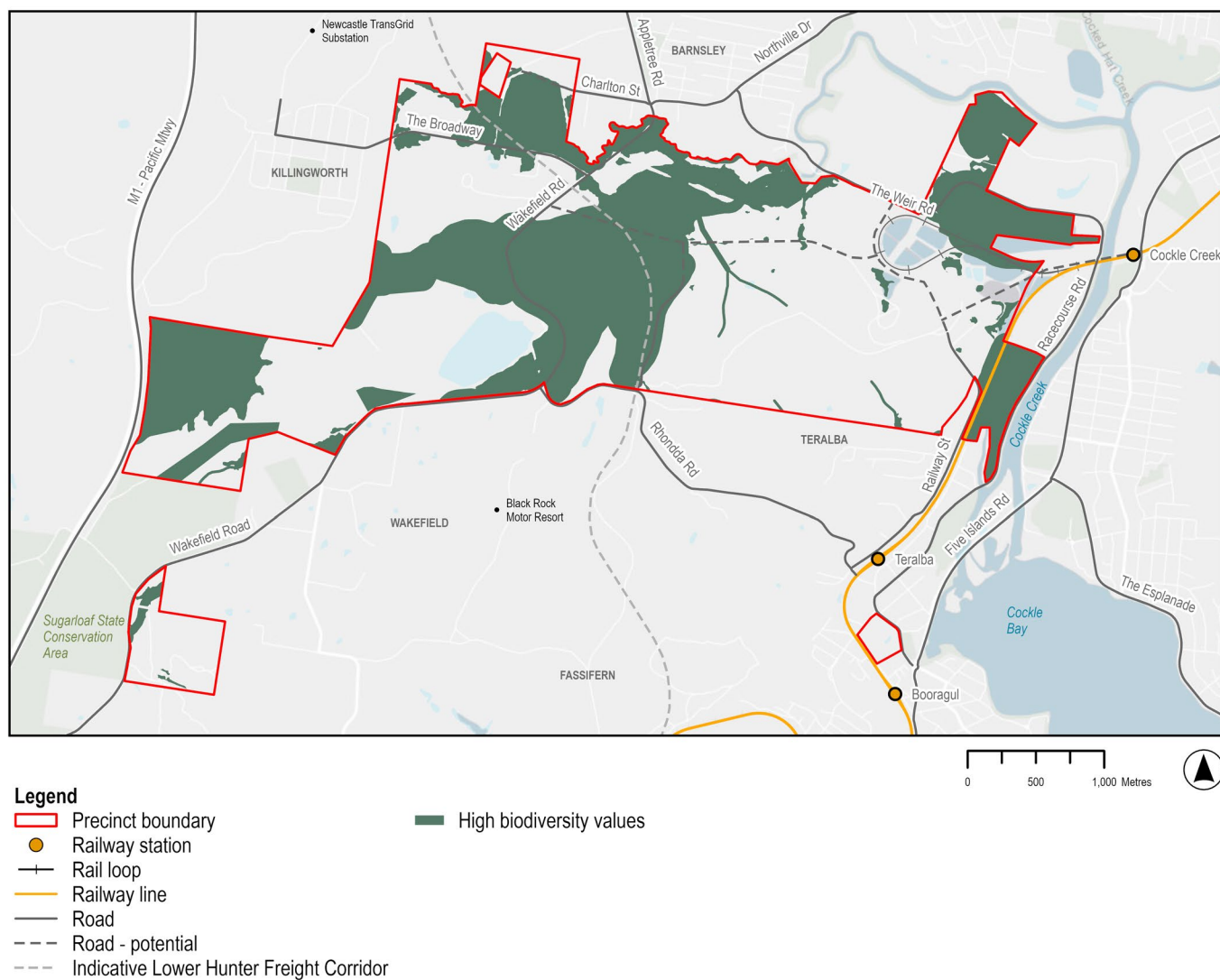
Aims

- Protect areas of high biodiversity values from development to avoid serious and irreversible impacts on threatened ecological communities and threatened species.
- Promote a development outcome that is consistent with the requirements of the avoid, minimise and offset hierarchy of applicable NSW and Commonwealth biodiversity legislation.
- Maintain the function of key linkages that facilitate wildlife movement within the site and the wider landscape.
- Identify and secure environmentally constrained land with an appropriate size and configuration to be protected under suitable conservation mechanisms.
- Protect wetlands, watercourses, and associated riparian corridors and aquatic habitats from development.
- Implement ecological restoration or rehabilitation measures on residual land or buffer zones to achieve a net positive biodiversity outcome.
- Apply ecologically sustainable development principles for future land use and development.
- Respond and adapt to future environmental changes to avoid adverse impacts on high biodiversity values.

Performance criteria

- A. Development footprints must be located to avoid and minimise impacts on areas of high biodiversity values that could result in a serious and irreversible impact on threatened ecological communities or threatened species.
- B. Development must avoid fragmentation of existing key wildlife corridors as identified in the draft Master Plan and must consider best-practice species specific guidelines for future corridor widths.
- C. Development must apply Stage 1 and Stage 2 and associated detailed requirements of the NSW Biodiversity Assessment Method for a future biodiversity assessment where entry into the NSW Biodiversity Offset Scheme is triggered.
- D. Appropriate buffers and setbacks in accordance with current applicable NSW and Commonwealth guidelines must be provided to areas of high biodiversity value to minimise indirect impacts and future degradation.
- E. Remnant native vegetation must be retained where practicable and vegetation removal must be avoided or minimised to maximise tree canopy cover within a development precinct.
- F. Development must incorporate suitable mitigation measures for restoration or rehabilitation of native vegetation communities and associated habitats within residual land or buffer zones that are enforced under a Vegetation Management Plan.
- G. Rehabilitation of native vegetation communities and landscape planting must prioritise locally endemic species and reflect the relevant plant community types.
- H. Development must avoid impacts to wetlands and associated buffer zones where the wetland provides important habitat for migratory shorebirds.
- I. Development must avoid riparian areas and should restore or rehabilitate the vegetated riparian zone where practicable.

Figure 28: High biodiversity values



Water cycle management

Water cycle management is a key consideration for future development within the Precinct. Diega Creek and Cockle Creek are the two key waterways located within the Precinct. Existing stormwater management within the Precinct is through natural flow paths within the undeveloped areas and via water management infrastructure across disturbed land. Water management infrastructure includes catchment dams, sediment dams, storage dams, spillways and drainage systems. Cockle Creek is the main receiver of stormwater runoff from the Precinct, receiving flows from upstream tributaries, natural flow paths and water directed from the disturbed areas via two licenced locations.

Aims

- Adopt an integrated water cycle management approach across the Precinct.
- Minimise impacts on surface water, groundwater and downstream receiving environments.
- Incorporate WSUD to improve water quality, resilience and place outcomes.
- Maintain natural drainage function, groundwater processes and recharge where feasible.
- Manage construction-phase erosion and sediment risks to protect sensitive receivers.

Performance criteria

- A. Development must achieve applicable stormwater quality targets, including pollutant load reduction and nutrient load limits relevant to the Lake Macquarie catchment.
- B. High-priority groundwater dependent ecosystems must be protected in accordance with the applicable policy framework.
- C. Development must demonstrate an integrated water cycle and WSUD response that minimises hydrological and water quality impacts on downstream sensitive receivers.
- D. WSUD measures must be designed to treat frequent storm events, including flows up to and including the three-month average recurrence interval (ARI) event.
- E. Development must maintain, to the extent practicable, natural flow paths, sub-catchment function and flow direction and must not rely solely on piped or heavily engineered stormwater systems where this would increase erosion, scour or downstream impacts.

Figure 29: Concept for integrating water sensitive urban design



7 Transport and connectivity

Road network, freight and access

Road network and access is influenced by the Precinct's legacy as former coal mine operations. Historic mining operations have shaped both the physical terrain and created a framework that continues to inform movement and access in the post mining context. Former haul routes and access tracks, originally designed to support extraction, processing and transport activities, establish a foundational spatial logic for the post mining land use movement network.

The approach to managing transport and mobility is multi-layered, combining infrastructure planning and staging through demand management and integration with land use outcomes. To facilitate efficient movement within and around the Precinct, the road network of sub-arterial, collector and local roads is structured as a clear hierarchy, linking developable land to the broader regional network to minimise reliance on the external local network.

Subject to further investigation, a potential new east-west crossing of Cockle Creek is identified as an opportunity to strengthen network resilience, unlock alternative freight access that bypasses constrained local roads (Racecourse Road, Railway Street and Rhondda Road) and improve connectivity to Cockle Creek Station.

Teralba is directly serviced by Teralba and Booragul Stations and forms a natural anchor for enhanced pedestrian and cycling connectivity to the wider Precinct via the Main Northern railway corridor and the future active transport network.

Aims

- Provide an efficient, legible and resilient road network that supports sequenced Precinct development.
- Support safe and efficient freight, workforce and visitor access to and within the Precinct.
- Align freight and access planning with the regional road and freight network, including future strategic corridors.
- Improve access across existing infrastructure barriers, including the Main Northern Railway.

- Integrate flood and bush fire resilience, emergency access and network safety into road planning.
- Deliver complete streets that support walking, cycling, buses and future transport upgrades.

Performance criteria

- A. The Precinct road network must connect to the regional road network and provide a legible hierarchy of sub-arterial, collector and local roads consistent with applicable design standards.
- B. Road network and intersection upgrades must be delivered ahead of, or concurrently with, the development stages they are intended to serve.
- C. Internal road networks must be delivered progressively to support sequenced development and reduce reliance on external local roads.
- D. Development applications must demonstrate, through detailed traffic assessment, that the surrounding network and relevant intersections can operate at an acceptable level of service for the proposed stage of development.
- E. Freight access routes and approved restricted access vehicle (RAV) connections must be established before freight-generating land uses commence.
- F. Heavy vehicle and freight movements must avoid residential streets unless no feasible alternative exists and acceptable mitigation is demonstrated.
- G. Former haul routes and access tracks should be prioritised for long-term heavy vehicle movements within and through the Precinct.
- H. A freight management plan must be prepared for freight-generating development and must address internal freight circulation, last-mile movements and community impact mitigation.
- I. Roads, intersections and bus facilities must be designed to provide safe pedestrian access and support public transport integration.
- J. Internal roads are to be located on developable land and, where practical, within existing

- disturbed corridors, unless an alternative alignment is required for safety or resilience reasons.
- K. Local road connections through the dam break inundation zone must remain contingent and must not be relied upon for development until the relevant dam safety and suitability requirements have been satisfied.
- L. The Precinct's road network, intersections, road reserves must be designed to support efficient and reliable public transport operations.
- M. Parking must be provided in accordance with an approved parking study and applicable guidelines.

Figure 30: Potential transport network (short-term)

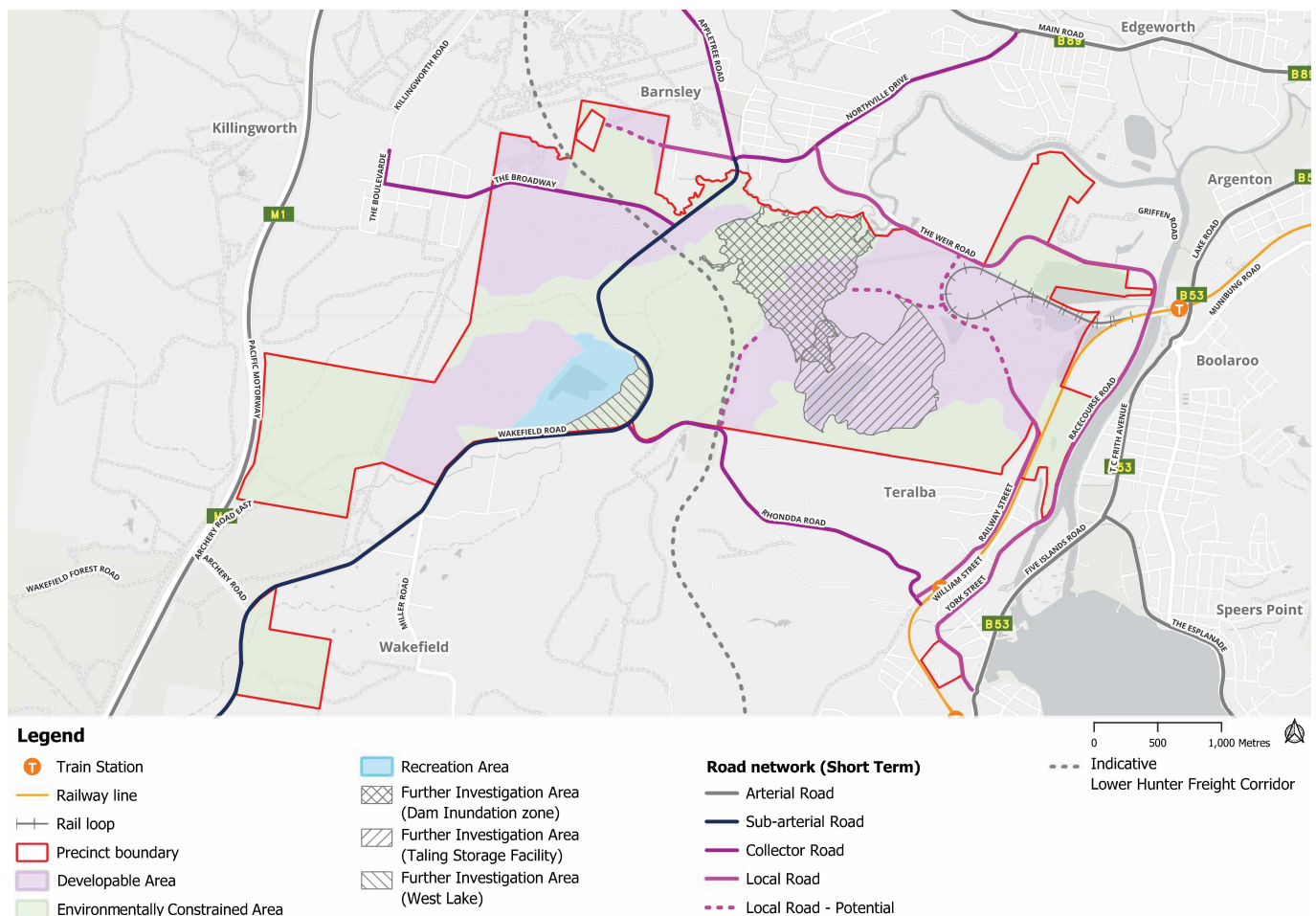


Figure 31: Potential transport network (long-term)

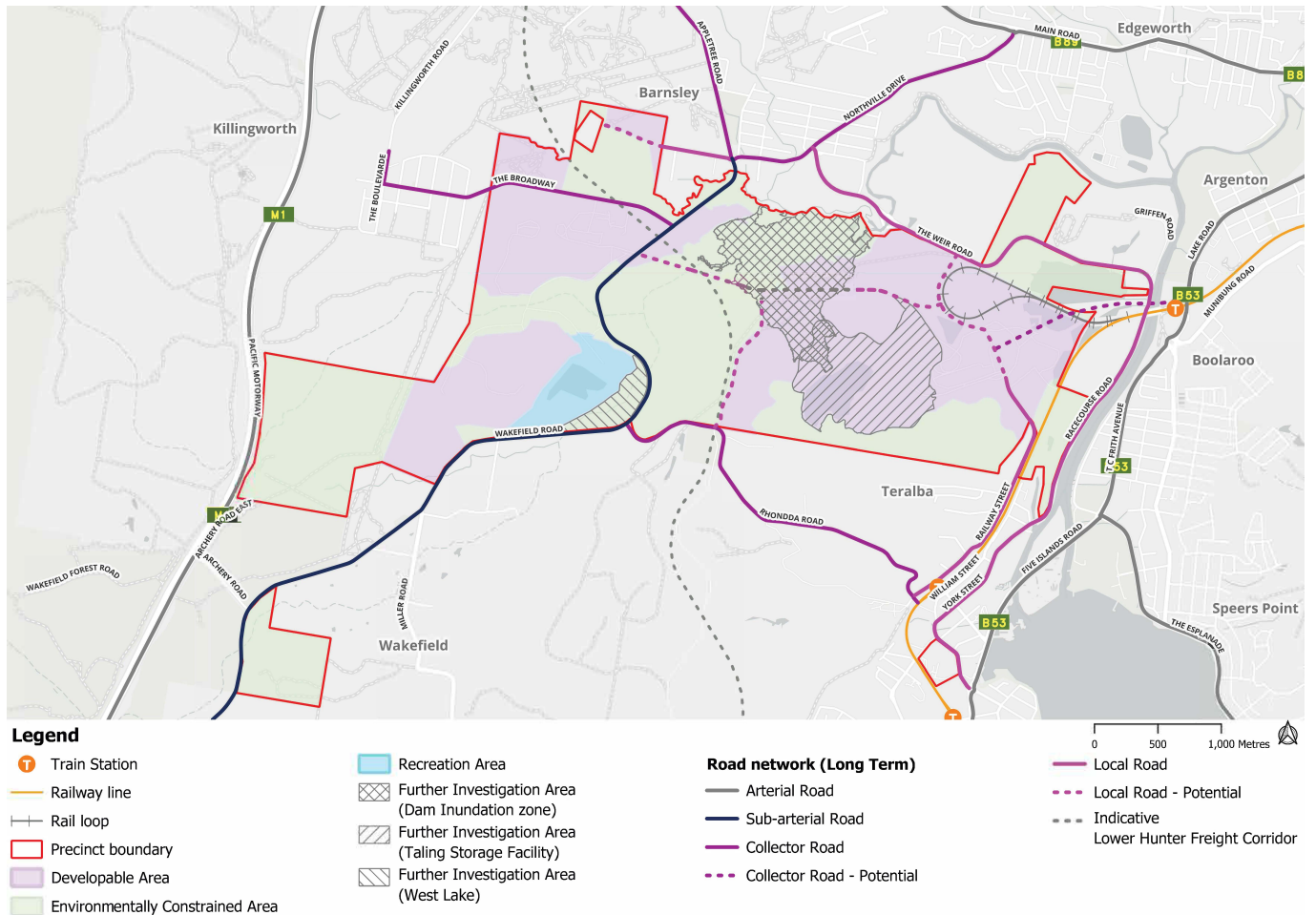


Figure 32: Future heavy vehicle network (short-term)

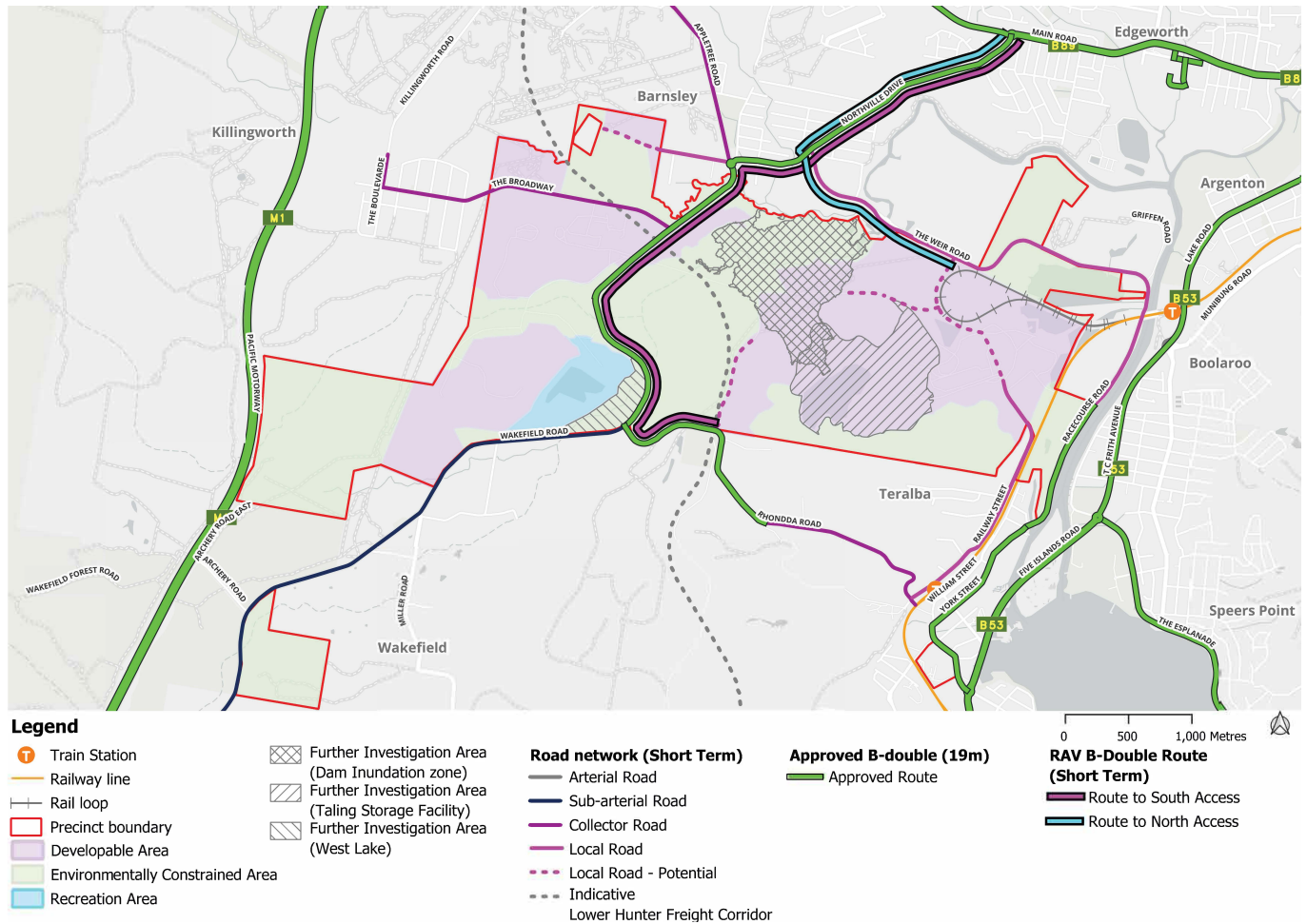
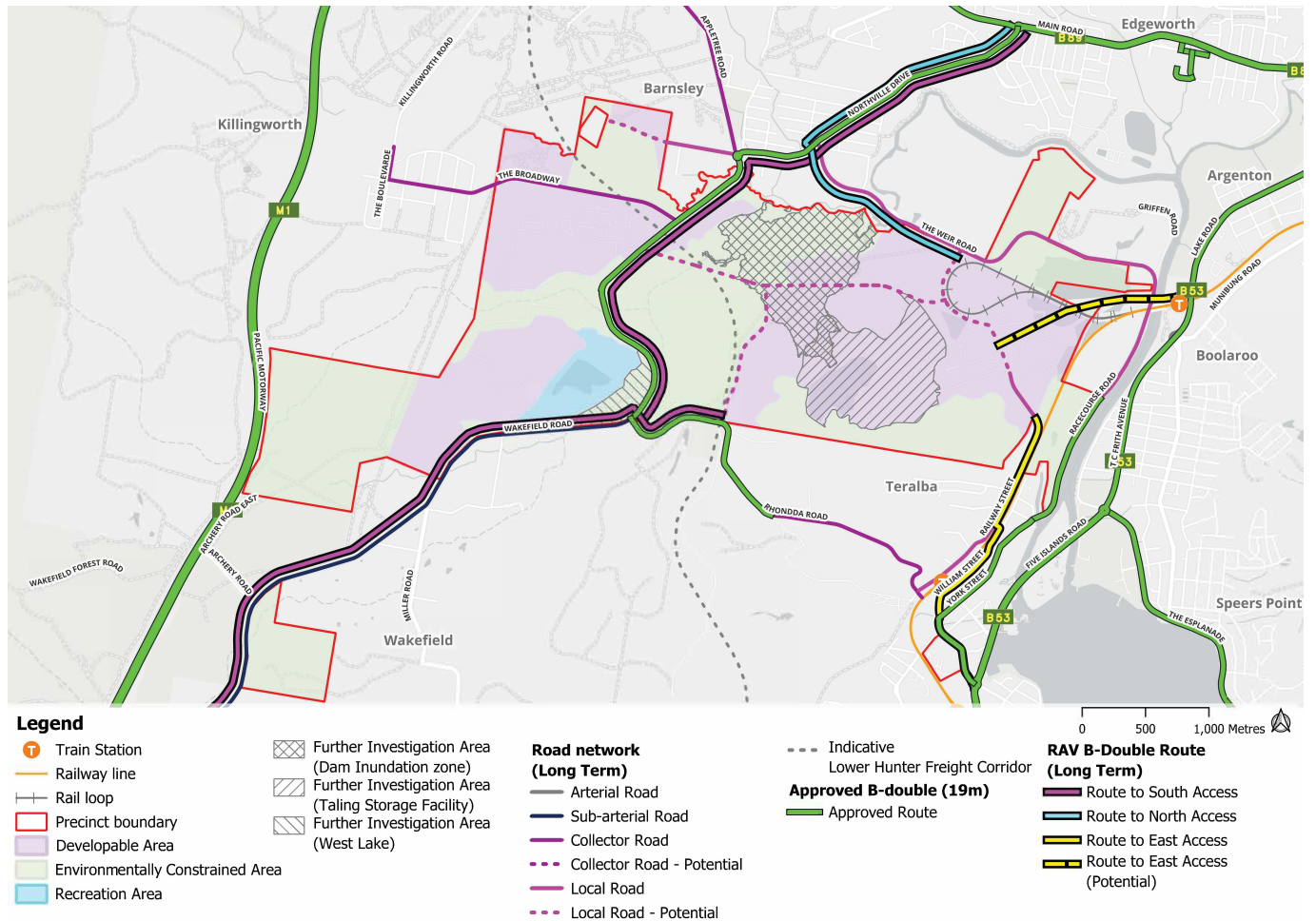


Figure 33: Future heavy vehicle network (long-term)



Active transport and connectivity

The Precinct provides a rare opportunity to plan a connected, safe and direct walking and cycling network, with preferred active transport alignments established with vehicle routes for optimal directness, legibility and separation from traffic. The active transport network will connect employment sub-precincts to Cockle Creek, Teralba and Booragul stations, surrounding residential communities, existing residential areas and the broader regional shared-path network.

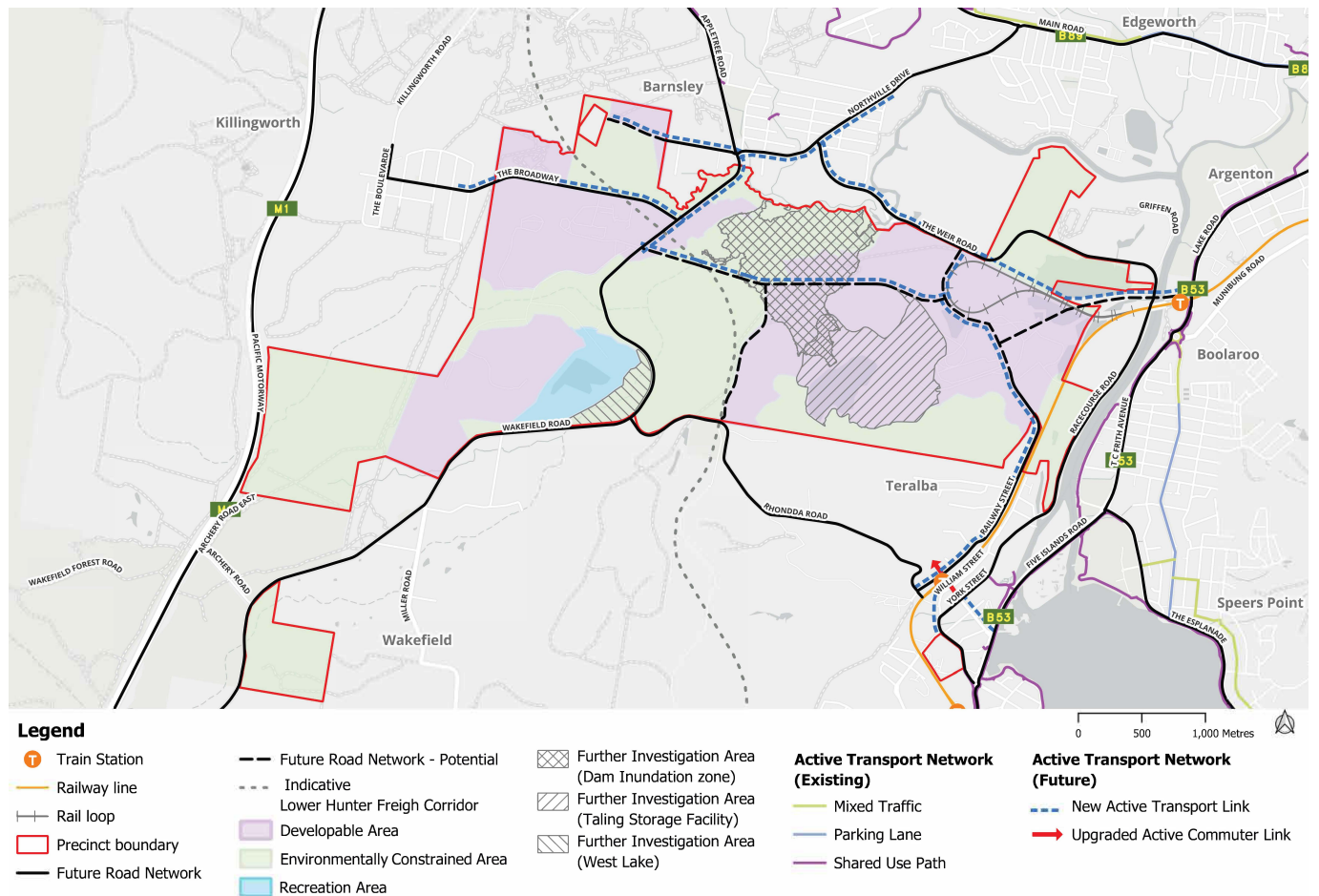
Aims

- Establish a connected and practical active transport network linking the Precinct to surrounding communities, centres and stations.
- Improve active transport access across key barriers including Five Islands Road and the Main Northern Railway.
- Integrate active transport with heritage, topographic and environmental constraints.
- Provide end-of-trip facilities and safe walking and cycling infrastructure as part of staged precinct delivery.

Performance criteria

- A. Pedestrian and cycling infrastructure must be delivered early to support activation and establish sustainable travel behaviour.
- B. Pedestrian and cycle connections must align with the indicative active transport network and connect sub-precincts, public transport nodes and open space.
- C. Active transport infrastructure must be designed in accordance with CPTED principles, universal access requirements and passive surveillance principles.
- D. Pedestrian and cycle links must be coordinated with road infrastructure delivery and incorporated into road reserves where practical.
- E. Active transport routes must connect to the wider regional shared path network at appropriate locations.
- F. Active transport infrastructure must respond sensitively to heritage streetscapes, steep terrain and environmental constraints.
- G. Development must provide end-of-trip facilities at a scale appropriate to the proposed use and intensity of development.

Figure 34: Existing and potential active transport network



8 Social infrastructure and community facilities

The Precinct is primarily employment-led and will not generate a significant residential population. Social infrastructure planning therefore focuses on supporting the daily needs of the Precinct's workforce and visitors, delivering community benefit to surrounding communities and ensuring that facilities are staged in alignment with development.

Community facilities

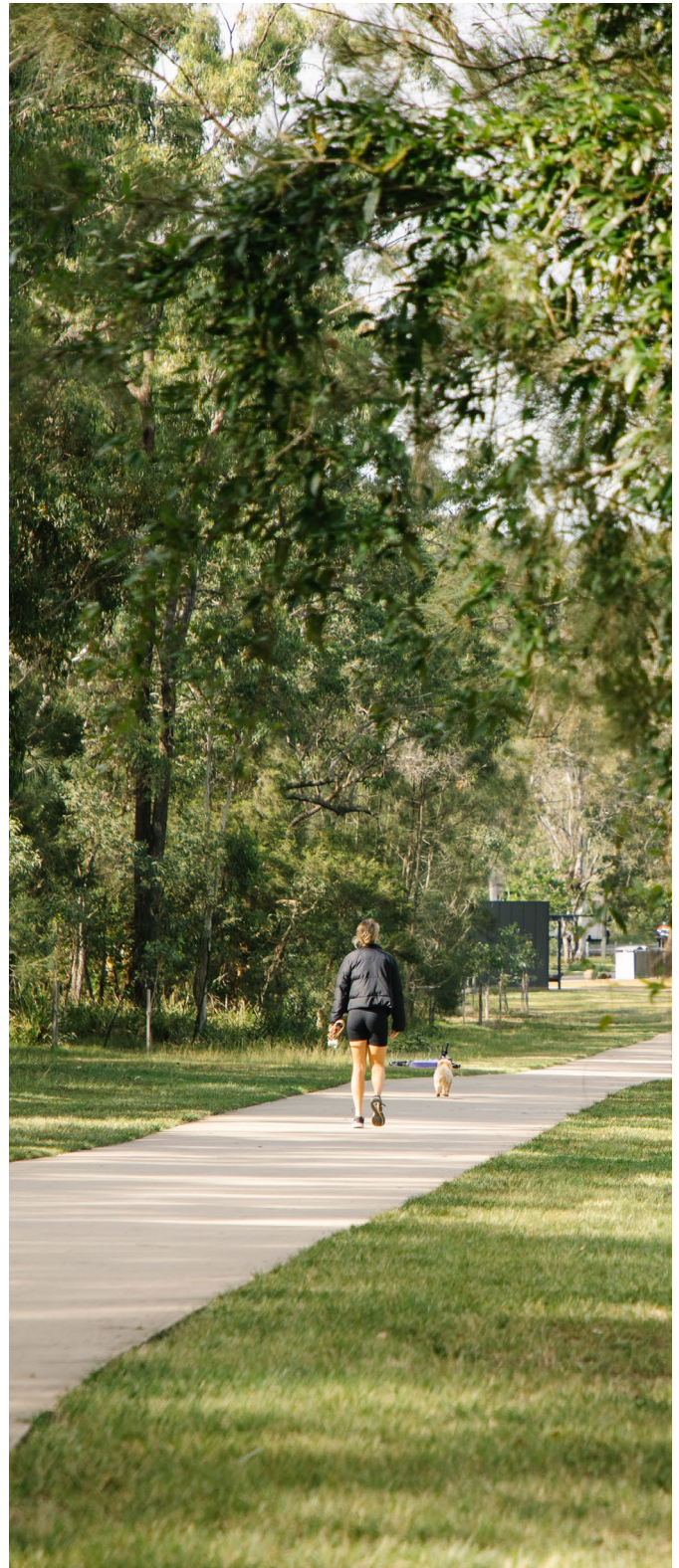
Aims

- Provide community facilities that support workers, visitors and surrounding communities.
- Locate community facilities and worker amenity hubs to activate key nodes and open space.
- Encourage adaptive reuse of suitable existing structures for community, cultural or training purposes.
- Make sure community facilities are accessible, appropriately sited and responsive to amenity and landscape context.
- Support opportunities for cultural and creative industries, including mining heritage interpretation and Aboriginal cultural enterprise.

Performance criteria

- A. Community facilities must provide flexible, publicly accessible multi-purpose space and must be co-located with employment areas or open space where feasible.
- B. Community facilities must be sited and designed to maintain solar access to adjoining public spaces, retain significant vegetation where feasible and provide safe, legible pedestrian access.
- C. Development that exceeds a workforce threshold identified through the social assessment must provide a centralised worker amenity hub incorporating food and beverage, convenience services and end-of-trip facilities.
- D. Community facilities must provide affordable or low-cost access options for community use, including not-for-profit and local organisations.

Figure 35: Different community facilities envisaged for the Precinct



Social infrastructure

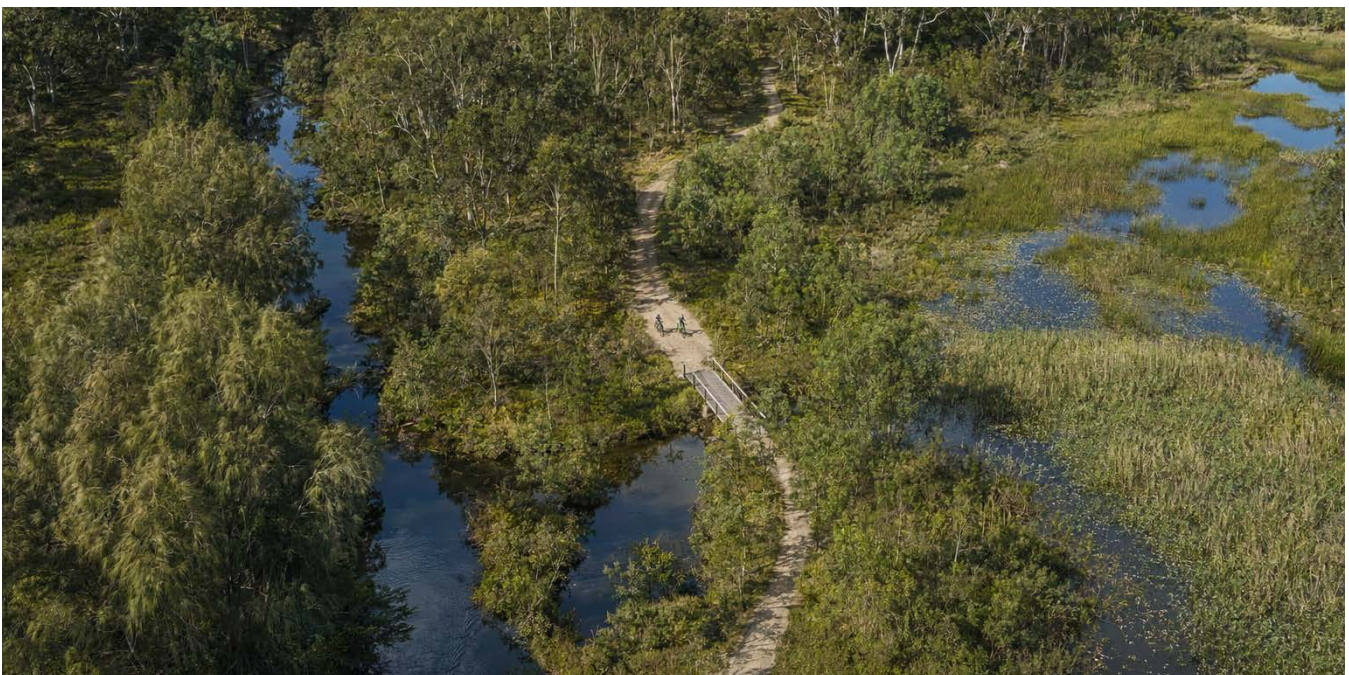
Aims

- Provide accessible open space, trails and public realm connected to the Precinct's natural, cultural and heritage assets.
- Deliver gathering spaces and active nodes that support inclusion, wellbeing and visitor amenity.
- Integrate Connecting with Country outcomes into public space, interpretation and trail infrastructure.
- Support workforce development through shared training and collaboration spaces.
- Deliver measurable community benefit and accessible recreation outcomes for surrounding communities.
- Support Aboriginal employment, enterprise and cultural participation as part of the Precinct's social infrastructure outcome.

Performance criteria

- A social infrastructure hierarchy and indicative sub-precinct staging triggers must be established, informed by the relevant social assessment and in consultation with Council.
- Social infrastructure must be aligned with employment areas, public open space and active transport connections where feasible.
- Development must provide opportunities or connections to active recreation, passive use and nature-based experiences that support health and wellbeing.
- Provide walking and cycling trails, trailheads and interpretive infrastructure within West Lake and the environmentally constrained area network that are connected to existing regional trail networks.
- Walking and cycling trails must be delivered as early activation infrastructure ahead of, or concurrently with, the first relevant development staging.
- Community and social infrastructure must achieve universal access outcomes, supporting mobility, safety and inclusion across all user groups.

Figure 36: Visualisation of potential social infrastructure envisaged for the Precinct



9 Sustainability and resilience

Tailings storage and dam break inundation zone

The former operations of the Precinct include a TSF, which is an engineered structure designed to safely store and manage rejected materials generated from coal mining operations, including solids and water. Closure of the TSF, including rehabilitation and capping is underway. The TSF will remain a declared dam until the structure is stabilised in accordance with the Precinct's mine closure, rehabilitation commitments and Dams Safety NSW approve the formal revocation of the declaration.

The draft Master Plan applies a staged risk-informed framework to manage hazards associated with the TSF and the potential for dam break downstream (dam break inundation zone) as described in Figure 9. This framework makes sure that future land use is compatible with the residual risk profile of the TSF over its remaining life and following de-declaration.

Dam break modelling has defined inundation extents, flow paths and level of hazard under a failure scenario. These outputs directly inform spatial planning by establishing compatible land use scenarios across the TSF areas that would enable flexibility over the period of rehabilitation and closure:

- Sensitive or high-occupancy land uses are avoided in the short to medium term within mapped inundation zones and buffers.
- Lower-intensity or transient uses may be considered in outer zones (such as passive and active recreation, storage facilities), subject to appropriate risk mitigation measures.
- Areas above the TSF could provide land uses which do not compromise the capping once rehabilitation is complete and a more stable landform is achieved over time.

New and existing infrastructure, such as roads that traverse the dam break inundation zone and have the potential to present risk, must be appropriately designed to make sure they are located above the modelled dam break inundation level whilst the TSF is a declared dam.

Aims

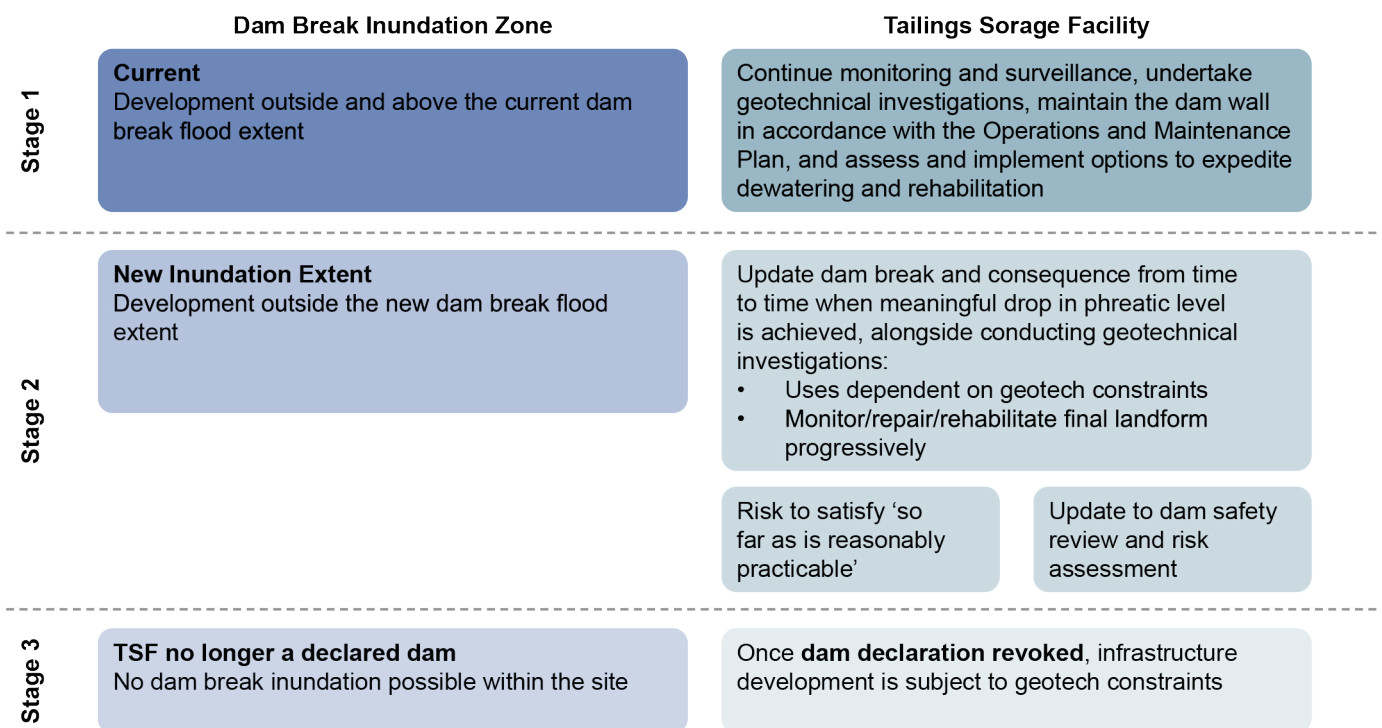
- Manage the TSF and dam break inundation zone through a risk-informed planning framework.
- Protect life, property and infrastructure from TSF-related hazards during rehabilitation and post-closure phases.
- Enable staged and adaptive land use outcomes as rehabilitation milestones are achieved and risk is reduced.
- Maintain dam safety, monitoring and maintenance until the TSF is revoked as a declared dam.

Performance criteria

- A. Development must not increase risk associated with the TSF or the dam break inundation zone and must not compromise rehabilitation, closure or long-term stability outcomes.
- B. Development staging must align with TSF rehabilitation progress and the current approved dam safety and consequence assessment framework.
- C. Land may only be considered for development outside an updated inundation extent where a revised dam break and consequence assessment has been approved by the relevant authority and demonstrates an acceptable reduction in risk.
- D. Development within, above or downstream of the TSF influence area must not intensify occupancy or use in a manner that increases societal risk while the TSF remains a declared dam.
- E. Any proposal within the dam break inundation zone must be supported by updated dam break consequence assessment, dam safety review and risk assessment proportionate to the proposed use.
- F. Roads and other infrastructure (including walking and cycling trails) that traverse the dam break inundation zone must be designed so as not to increase risk to life and, where required, must be located above the modelled inundation level and designed to withstand flood impacts.

- G. Development is to enable access for ongoing management of the TSF.
- H. Utility infrastructure proposed downstream of the TSF must demonstrate that it will not adversely affect TSF stability, operation or rehabilitation.
- I. A minimum 10-metre buffer must be maintained around the TSF footprint, unless a more conservative requirement is established through detailed engineering assessment.
- J. Following revocation of the TSF dam declaration, any works within the TSF must be accompanied by suitably detailed engineering assessment certifying that the works will not adversely affect short or long-term stability.
- Post-closure land uses within the TSF area must respond to residual geotechnical and settlement constraints and remain compatible with the long-term performance of the landform.

Figure 37: Recommendations to support staged development



N.B. No development on dam wall and within 10m of dam footprint until de-declared

Flooding and risk management

Flooding is a key consideration for future development within the Precinct due to the presence of Cockle Creek, Diega Creek, Burkes Creek and associated floodplain areas. Flood studies indicate that flood behaviour within and adjacent to the Precinct is primarily associated with creek flooding, floodplain inundation and localised drainage constraints in low-lying areas. Flood behaviour is influenced by the Cockle Creek system, associated tributaries and downstream hydraulic controls including the Barnsley Weir and Lake Macquarie water levels.

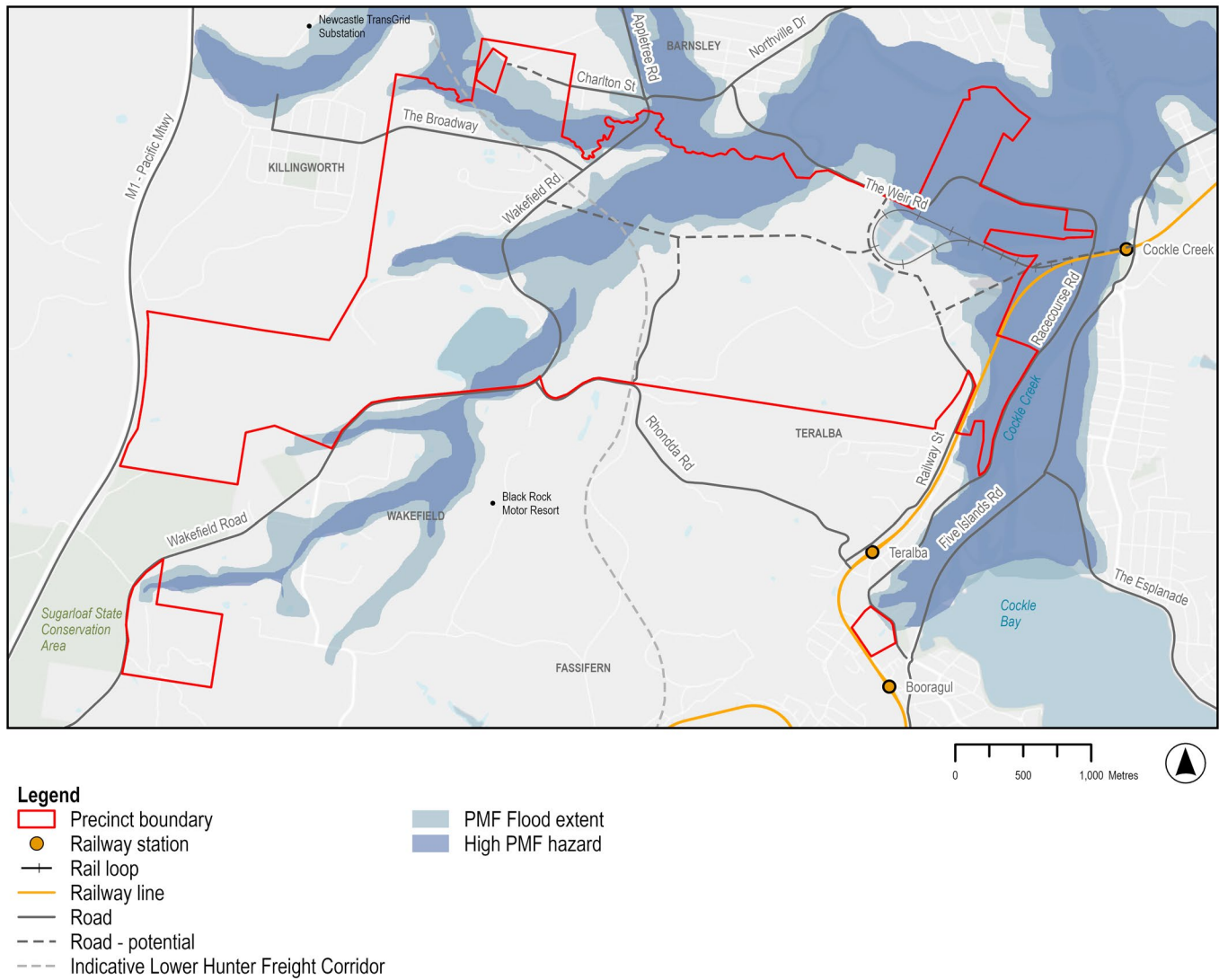
Aims

- Direct development away from areas of higher flood risk and drainage constraint where practicable.
- Protect flood conveyance, flood storage and waterway function.
- Make sure development is compatible with flood hazard and does not increase flood risk elsewhere.
- Integrate flood resilience, emergency access and isolation risk into precinct planning.
- Require detailed flood, drainage and emergency management assessment for flood-affected land.

Performance criteria

- A. Development must have regard to the flood-affected areas identified in Figure 38 and any subsequent detailed flood investigations.
- B. Flood planning levels, including freeboard, must align with Council requirements and incorporate climate change allowances to year 2100.
- C. Development must demonstrate that it will not unreasonably increase flood levels, extents, velocities or hazard on surrounding land.
- D. Development on flood-affected or drainage-constrained land must demonstrate that flood risk is acceptable for the proposed development type or has been mitigated to an acceptable level.
- E. Future road, access and movement planning must respond to flood risk, flood immunity and isolation risk, having regard to the required level of service.
- F. Flood-constrained land is not intended for intensive, evacuation-sensitive or critical development unless supported by detailed flood, emergency management and risk assessment.
- G. Development applications on, or affected by, flood-prone land must be supported by detailed flood, drainage and emergency management investigations.

Figure 38: Flood-prone land



Bush fire resilience

Bush fire resilience is a key design parameter shaping the structure and long-term function of the precinct. The draft Master Plan adopts a precinct approach to bush fire planning, responding to the site's classification as bush fire prone land and the complex interaction between vegetation, topography and development. The draft Master Plan prioritises consolidated development footprints, clearly defined edges and integrated interface treatments that support both bush fire resilience and biodiversity outcomes.

Aims

- Protect life and property from bush fire risk while responding to development potential, constraints, operational requirements, rehabilitation outcomes and the protection of biodiversity and habitat corridors.
- Enable ongoing land management practices that concurrently support bush fire resilience, protection of high value biodiversity, and the long-term health and diversity of rehabilitated and retained landscapes.
- Integrate road hierarchy, access, egress and emergency response to support operations, firefighting and evacuation.
- Enable suppression, staged responses and long-term resilient precinct activation through strategic separation, managed land and services, including compliant water supply.

Performance criteria

- A. Development must be located only where an integrated and appropriate level of bush fire protection can be achieved having regard to high value biodiversity, vegetation, topography, fire run potential, rehabilitation outcomes and access.
- B. Higher-vulnerability and higher-consequence land uses must be located in areas of comparatively lower bush fire exposure, with increased separation from unmanaged vegetation and hazard areas.
- C. Development must not create isolated pockets of development, rely on a single access route, or result in layouts that increase bushland-urban interface exposure or limit emergency response capability.

- D. Development interfaces exposed to unmanaged bush fire hazard must be minimised in both length and complexity, with preference given to consolidated development footprints and regular, defensible edges.
- E. Vegetation corridors that may facilitate bush fire spread towards development areas must be appropriately configured and managed.
- F. Rehabilitation and vegetation management outcomes must be integrated into long-term bush fire management arrangements, making sure that future vegetation growth does not compromise asset protection zones or interface performance over time.
- G. Development must be designed to minimise the total extent of bushland-urban interface and avoid fragmented or elongated hazard edges.
- H. All interfaces must incorporate clearly defined and continuous edge treatments that support both bushfire protection and biodiversity outcomes.

Geotechnical

The Precinct is characterised by a highly variable post mining landscape, where geological conditions, legacy mining impacts and anthropogenic landforms generate a range of geotechnical constraints influencing development feasibility. Development will require a risk-based and sequenced approach with consideration of landform suitability and engineered earthworks solutions, including cut and fill, batter stabilisation and retaining structures. Site-specific geotechnical investigation would be required to confirm conditions and inform design.

Key geotechnical considerations include:

- Possible subsidence from former mining operations
- Presence of soft and potentially saturated alluvial sediments around waterways
- Steep topography
- Underlying coal seams
- Potential tuffaceous materials

- Areas affected by historic filling activities
- Acid sulfate soils (ASS)
- The stability of land associated with the TSF (Refer to **Section 9 Sustainability and resilience on page 51**).

Aims

- Make sure development responds appropriately to geotechnical and possible mine subsidence constraints.
- Prioritise development in areas with comparatively lower geotechnical complexity.
- Require development to be informed by fit-for-purpose geotechnical investigation consistent with the proposed land use.

Performance criteria

- Development must be informed by detailed geotechnical assessment that addresses relevant historical mine workings, settlement, rehabilitation, slope stability and ground conditions.
- Development, including earthworks, must be sited, designed and constructed to appropriately manage identified geotechnical risks.
- Development must not penetrate, disturb or compromise the integrity of the TSF cap.
- All works within or adjacent to the TSF must demonstrate protection of the cap and be supported by detailed geotechnical assessment and design consistent with dam safety and rehabilitation requirements.
- Development must demonstrate that site slopes are suitable for the intended land use and where slopes are not suitable, development must be supported by an engineered landform response.
- Development must be supported by an earthworks strategy that:
 - Minimises excessive cut and fill
 - Addresses batter stability and long-term erosion risk
 - Incorporates retaining structures where required
 - Demonstrates safe and feasible construction outcomes.

- Acid sulfate soils investigations must be undertaken where it is known or likely to occur.
- Slope stability investigations must be undertaken where topography or landform conditions indicate potential instability.

Contamination

The Precinct comprises a former mining and industrial landscape where contamination conditions reflect historical land use and rehabilitation activities. Contamination has been investigated to support mine closure, with a staged program of assessment and remediation ongoing. Following completion of rehabilitation works, contamination associated with mining activities will be remediated by the current landowner to a standard that enables future development, subject to verification through site-specific investigation at the development application stage. The draft Master Plan adopts a risk-based approach to make sure that any residual contamination is appropriately managed and that land is suitable for its intended post mining use.

Aims

- Make sure contamination is identified, assessed and managed in a manner proportionate to the proposed land use.
- Provide clarity on contamination constraints, investigation requirements and remediation pathways to support staged development.
- Recognise that contamination associated with historical mining activities will be remediated by the current landowner as part of rehabilitation prior to, or in parallel with, development.

Performance criteria

- Development applications must be accompanied by contamination assessment sufficient to demonstrate that the land is suitable, or can be made suitable, for the proposed use in accordance with the applicable statutory framework.
- Contamination assessment, remediation and management must be proportionate to the sensitivity of the proposed land use and the level of residual risk following rehabilitation.

- C. Contamination mitigation and remediation measures must be integrated with flood, drainage, geotechnical and water management strategies to achieve coordinated human health, environmental and land stability outcomes.

Air quality and odour

Aims

- Protect air quality and amenity for workers, visitors and adjacent residents.
- Avoid or minimise adverse air quality and odour impacts on surrounding sensitive receivers and environmental values.
- Make sure development is sited, designed and operated to manage cumulative emissions and odour impacts.
- Require air quality and odour management appropriate to each stage of development.

Performance criteria

- A. Development must avoid or minimise air pollutant emissions and odour impacts so that acceptable human health and amenity outcomes are achieved for workers, visitors and adjacent residents.
- B. Significant air emission sources and sensitive receptors must be separated through appropriate siting, design, buffering and operational controls.
- C. Development that may generate material air quality or odour impacts, or that is proposed in a sensitive interface area, must be supported by an air quality and odour assessment proportionate to the proposed use.
- D. Development must be designed to respond to existing and ongoing air quality and odour generated by TSF management and rehabilitation, making sure that land use conflicts are avoided and ongoing TSF operations are not constrained.

Noise and vibration

Aims

- Manage noise and vibration to achieve appropriate amenity outcomes for surrounding receivers and future occupants of the Precinct.
- Provide appropriate separation and mitigation between industry, transport infrastructure and sensitive land uses.
- Make sure development within the Precinct manages cumulative construction and operational traffic noise impacts on adjacent residential land uses outside the Precinct, including those arising from staged development.

Performance criteria

- A. Development applications for noise-sensitive uses, or for uses with the potential to generate material noise or vibration impacts, must be supported by a noise and vibration assessment that identifies likely impacts and appropriate mitigation measures.
- B. Noise-sensitive development must be designed to minimise exposure through site layout, building orientation, buffering and acoustic treatment.
- C. Road upgrades and related transport infrastructure must demonstrate that traffic noise impacts on sensitive receivers are minimised through design and mitigation measures consistent with relevant guidelines.
- D. Noise vibration impacts associated with staged precinct development, including cumulative construction and operational impacts, must be identified and managed to achieve acceptable noise levels under the NSW Industrial Noise Policy.
- E. Development must be designed to respond to existing and ongoing noise and vibration generated by TSF management and rehabilitation, ensuring that land use conflicts are avoided and ongoing TSF operations are not constrained.

10 Infrastructure, utilities and servicing

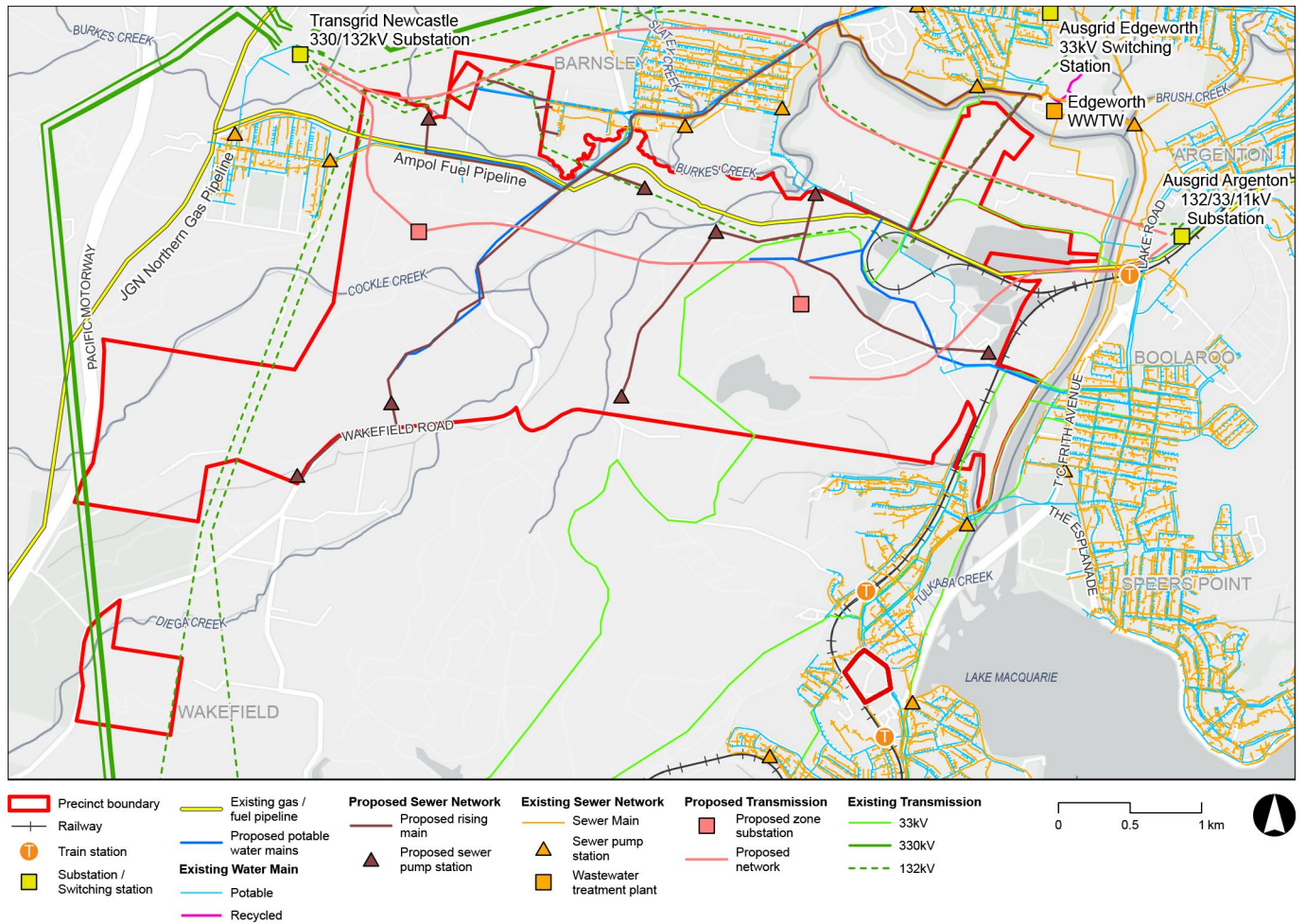
Aims

- Deliver infrastructure, utilities and servicing that are safe, reliable, efficient and coordinated with precinct staging.
 - Provide that infrastructure delivery supports long-term flexibility, resilience and orderly development.
 - Integrate utility planning with public realm, biodiversity values and amenity.
 - Support circular-economy, decentralised energy and future-ready servicing solutions where feasible.
 - Minimise disruption through coordinated corridor planning and a 'dig once' approach.
 - Support resource recovery, blue-green infrastructure and climate resilience in infrastructure delivery.
- E. Development must respond to the location, capacity, protection requirements and upgrade pathways of existing and future utility infrastructure.
 - F. Sensitive uses, high-occupancy uses, critical emergency infrastructure and difficult-to-evacuate populations must not be located near the fuel pipeline unless detailed risk assessment demonstrates an acceptable outcome.
 - G. Development must provide telecommunications infrastructure of sufficient capacity to support intended future use.
 - H. Development must not preclude opportunities for circular-economy servicing, decentralised energy systems or other sustainable infrastructure solutions.
 - I. Water cycle management infrastructure, including erosion and sediment controls, WSUD and detention measures, must be delivered in sequence with development to protect waterways and ecosystems.
 - J. Utility infrastructure must be designed to be resilient to climate change, including flood, bush fire, heat and service continuity risks relevant to the proposed development and location.
 - K. Infrastructure, utilities and servicing must be located, designed and managed to avoid or minimise impacts on biodiversity values.

Performance criteria

- A. Utility infrastructure must be generally consistent with the existing and potential utilities servicing described in Figure 39.
 - B. Utility infrastructure must be sequenced and coordinated with development uptake and precinct sequencing to avoid inefficient servicing patterns and out-of-sequence development, unless on-site servicing is demonstrated to be practical and feasible.
 - C. Development must demonstrate access to wastewater servicing, whether through a new network, Hunter Water or an approved alternative servicing solution appropriate to the proposed use and stage.
 - D. Utility connections must provide sufficient capacity and conduit allowance to accommodate the intended use and reasonably foreseeable changes in demand, technology or land use.
- E. Development must respond to the location, capacity, protection requirements and upgrade pathways of existing and future utility infrastructure.
 - F. Sensitive uses, high-occupancy uses, critical emergency infrastructure and difficult-to-evacuate populations must not be located near the fuel pipeline unless detailed risk assessment demonstrates an acceptable outcome.
 - G. Development must provide telecommunications infrastructure of sufficient capacity to support intended future use.
 - H. Development must not preclude opportunities for circular-economy servicing, decentralised energy systems or other sustainable infrastructure solutions.
 - I. Water cycle management infrastructure, including erosion and sediment controls, WSUD and detention measures, must be delivered in sequence with development to protect waterways and ecosystems.
 - J. Utility infrastructure must be designed to be resilient to climate change, including flood, bush fire, heat and service continuity risks relevant to the proposed development and location.
 - K. Infrastructure, utilities and servicing must be located, designed and managed to avoid or minimise impacts on biodiversity values.

Figure 39: Existing and potential utilities servicing



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